



Imported intermediates, technological capabilities and exports: Evidence from Brazilian firm-level data

Caio Torres Mazzi^{*}, Neil Foster-McGregor

UNU-MERIT, Boschstraat 24, Maastricht 6211AX, the Netherlands

ARTICLE INFO

JEL codes:

F14
O30

Keywords:

Imported intermediates
Exports
Productivity
Innovation
Technological capabilities

ABSTRACT

This paper explores how internal technological capabilities influence the relationship between imported inputs and the export performance of firms. We apply threshold regression techniques to a representative dataset of Brazilian firms and find a strong positive influence of innovation skills on the relationship between imported intermediates and export revenues. Complementarities between capabilities and importing are found only for high-quality imports and are stronger for exports of products with a higher scope for quality differentiation. We also observe that technological capabilities are directly correlated with export performance, confirming the view that innovation positively influences firms' international competitiveness. However, this relationship is not found to be significant for firms that export products with a low scope for quality differentiation. Overall, our results suggest that technological capabilities and the quality of imported inputs not only benefit firms directly but also complement each other in enhancing export competitiveness.

1. Introduction

A large theoretical and empirical literature has emerged addressing how access to imported inputs affects the performance of domestic firms. Initial studies demonstrated that imports not only increased competitive pressure on local markets, forcing firms to become more efficient, but could also influence their performance by expanding the availability of foreign intermediates (Amiti & Konings, 2007; Kasahara & Rodrigue, 2008; Topalova & Khandelwal, 2011). The literature that followed quickly indicated that the channels that mediate the effect of imported inputs on firms' performance relate not only to reductions in the cost of inputs but also to the superior embodied technology, quality and variety of many imported intermediates (Bas & Strauss-Kahn, 2014; Bas & Strauss-Kahn, 2015; Halpern et al, 2015; Kasahara & Lapham, 2013; Goldberg et al, 2010).

The international trade literature suggests that if access to imported intermediates increases firm performance it will also positively impact their exports. More productive firms 'self-select' into exporting, as they are the ones able to afford the relatively high fixed-costs of entering foreign markets (Clerides, Lach, & Tybout, 1998; Melitz, 2003; Wagner, 2007). However, the evidence also indicates that imported intermediates facilitate firms' efforts to upgrade their product portfolios and meet the requirements of foreign buyers, thus increasing demand in

foreign markets and boosting the intensive and extensive margins of exports. Empirical studies have shown that there is a strong connection between imports and both the variety and the quality of exports at the firm level. (Bas & Strauss-kahn, 2015; Bas & Strauss-Kahn, 2014; Fan, Li, & Yeaple, 2015; Feng, Li, & Swenson, 2016; Kugler & Verhoogen, 2009a, 2012; Manova & Zhang, 2012; Wagner, 2016).

In this context, the focus of this paper is on the role of firms' technological capabilities and the influence of these capabilities on the relationship between imported inputs and the export performance of firms. We test how firms' technological capabilities condition the correlation between imported inputs and export performance by applying threshold regression techniques to the most representative dataset of Brazilian firms. Our empirical analysis is oriented by first revisiting the Kugler & Verhoogen (2012) model that studies the role of firm heterogeneity in explaining the relationship between the quality of imports and exports. As with other related works in the international trade literature (Bastos, Silva, & Verhoogen, 2018; Demir, 2013; Hallak & Sivadasan, 2013; Johnson, 2011; Manova & Yu, 2017), this model develops a framework in which high-capability firms have a comparative advantage in producing high-quality goods. Moreover, firms depend on complementarities between internal capabilities and the quality of their inputs to increase the quality of their own final products. This description of the quality production process summarizes the most important

^{*} Corresponding author.

E-mail addresses: torres@merit.unu.edu (C. Torres Mazzi), foster@merit.unu.edu (N. Foster-McGregor).

<https://doi.org/10.1016/j.respol.2020.104141>

Received 26 August 2019; Received in revised form 20 July 2020; Accepted 6 October 2020

Available online 2 November 2020

0048-7333/© 2020 Elsevier B.V. All rights reserved.

aspect of our empirical results, i.e. that firms' export performance depends on innovation-related factors, which in turn depend on their internal technological capabilities and on external technology obtained through imported inputs.

In the model, consumers obtain utility not only from the quantity but also from the quality of the products they consume. Firms decide both prices and the quality level of the varieties they offer. However, higher-quality inputs are more expensive and increase the price of final products, creating a trade-off for producers: in order to upgrade quality, it is necessary to increase production costs, which might translate into more expensive final products. In equilibrium, more capable firms will tend to export more, use higher quality inputs and produce higher quality goods, with this relationship becoming stronger in markets with a higher scope for quality differentiation.

We concentrate on firms' export revenues as a measure of export performance but also look at the number of country-product pairs exported. These approaches connect our empirical framework also to studies that investigate the effect of imported intermediates on firms' overall performance and variety in exports. [Goldberg et al \(2010\)](#) provide evidence that a wider variety of imported intermediates had a sizable effect on the scope of products supplied by Indian firms in the internal market. Several studies find compatible evidence also for external markets, demonstrating a significant impact of input tariff reductions on the use of imported intermediates and their impact on firms' overall export performance ([Aristei et al, 2013](#); [Bas, 2012](#); [Edwards et al, 2018](#)), as well as the connection between input variety and the number of export destinations and product varieties ([Bas & Strauss-Kahn, 2014](#); [Manova & Zhang, 2012](#); [Wagner, 2016](#)). Such results are related to the existence of imperfect substitution between inputs, which – in the presence of increased access to imported intermediates and input diversity – leads to more efficient use of production factors.

Our analysis is also connected to the literature that focuses on innovation and “technology gaps”. The relationship between imports and innovation has long been recognized in these literatures in the context of capital goods ([Abramovitz, 1989](#); [Gerschenkron, 1962](#)) and more recently in the context of imported intermediates ([Castellani & Fassinio, 2019](#)), where the latter shows the positive effect of new imported products for firms that start exporting a new product variety for the first time. Our study complements this work by demonstrating that the effect of imports is conditioned by the presence of innovation capabilities within the firm. This is in line with the notion that imported intermediates have a strong impact on firms' technological choices, therefore suggesting that the extent to which firms will benefit from the higher quality/technology of foreign intermediates – or simply from the increased variety of inputs – in terms of innovation must also depend on their capacity to properly understand and manage technology. In that context, many authors have previously emphasized the importance of technological capabilities, i.e. the skills, knowledge and institutional structures necessary for firms to manage technology and develop technological change ([Bell & Pavitt, 1993](#); [Lall, 1992](#); [Morrison et al, 2008](#)), and the dual role of R&D and human capital for both generating and absorbing external sources of knowledge ([Aw et al, 2007](#); [Cohen & Levinthal, 1989](#)).

Most of the empirical literature exploring the role of quality in exports and imports assumes that capability is a completely unobserved variable that can only be indirectly inferred from firms' observed productivity ([Bas & Strauss-Kahn, 2015](#); [Fan et al., 2015](#); [Kugler & Verhoogen, 2012](#); [Manova & Yu, 2017](#)). Instead, we chose a different path and assume that firms' technological capabilities can be, to a meaningful extent, observed directly in the data through specific variables. This approach has the important advantage of allowing us to create a simple empirical framework capable of testing both the direct effect of capabilities on firms' export performance and the indirect effect related to complementarities between capabilities and inputs. That is not to say that our proxy – or any quantitative variable for that matter – is able to reflect the full range of firm-level capabilities. Nevertheless, following

the conceptualization of [Bell and Pavitt \(1993, 1995\)](#) and [Lall \(1992\)](#), among others, we take advantage of rich Brazilian labour market administrative information and use the functions performed by firms' employees to construct a proxy for the presence and intensity of ‘intermediate’ and ‘advanced’ complexity technological capabilities ([Lall, 1992](#), p. 167), which closely relate to the innovation process.

Finally, our work is also connected to studies on the relationship between innovation and export competitiveness. [Flach \(2016\)](#) directly associates quality improvements in firms' exports to product innovations, while other studies have shown that technology influences exports through channels that consistently remain important even after controlling for self-selection based on productivity. This is usually the case for R&D investments ([Aw et al, 2007](#); [Barrios et al, 2003](#); [Caldera, 2010](#)) and for innovation outputs, such as process and product innovations ([Becker & Egger, 2013](#); [Caldera, 2010](#); [Cassiman et al, 2010](#); [Damijan & Kostevc, 2015](#); [Dosi et al, 2015](#)). Since the introduction of new products will likely be preceded by investments in the accumulation of technological capabilities, the latter should also have a direct effect on firms' export performance. We recognize this possibility in the empirical analysis by allowing technological capabilities to directly impact upon exports.

Our main results are as follows: first, we observe that more capable firms benefit more from high-quality imported intermediates in terms of their export performance; second, the type of exported product conditions complementarities between input quality and firm capability, with the export of customized products – which offer greater scope for differentiation – significantly intensifying these complementarities compared to non-customized products; third, we observe that technological capabilities are directly correlated with export performance, with this relationship being significantly stronger for products with a higher scope for quality differentiation. Overall, our results indicate the existence of complementarities between imported inputs and firms' technological capabilities, with these complementarities driven by the import of higher quality intermediates.

We contribute to the literature in three main ways. First, to the best of our knowledge, this is the first framework providing evidence of complementarities between the quality of imported inputs and firms' technological capabilities by using direct proxies for both variables, as explained above. Second, we revisit the relationship between technological capabilities and export performance and qualify previous results by highlighting the influence of exported product types in this relationship. Third, we complement recent evidence ([Castellani & Fassinio, 2019](#)) on the differential effects of imported intermediates and imported capital goods, demonstrating how capabilities increase the importance of the former more intensively. Our results demonstrate that the benefits of trade tend to be asymmetric and much more intense for a minority of more capable firms that can take full advantage of accessing high-quality inputs.

The remainder of this paper is organized as follows. In [Section 2](#), we develop the theoretical framework that orients our empirical analysis based on a version of the main model of this literature. In [Section 3](#), we explain our empirical methodology, with data sources and the construction of variables being presented in [Section 4](#). In [Section 5](#) we present our empirical results, while conclusions and areas for future research are discussed in [Section 6](#).

2. Theoretical framework

2.1. The model set-up

We illustrate our main propositions by relying on the model of [Kugler & Verhoogen \(2012\)](#) and the more recent version developed in [Bastos et al \(2018\)](#), which we slightly alter to maintain the original monopolistic competition environment. We inherit all simplifying assumptions of this model, including the “partial-equilibrium” set-up in which the firm's decision to enter each market is assumed as given. There are K countries

and each firm supplies a single product in each market but with different quality levels. There is a separate production line for each market, which allows firms to choose the quality level of intermediates according to the chosen level of product quality. We index each product and country with subscripts i and k , respectively. Since each firm produces a single product, firms can also be indexed with subscript i , whereas product lines or product-quality pairs are identified by subscripts ik .

Consumer willingness to consume quality is strictly positive but differs by country. A common assumption of the literature is that consumers in wealthier countries are more willing to pay for quality than in poorer nations. The representative consumer maximizes a standard CES utility function augmented by quality and a parameter $d_k > 0$ determining consumers' willingness to pay for quality in each country k :

$$U_k = \left(\int_{i \in \Omega} (q_{ik}^{d_k} x_{ik})^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}} \quad (1)$$

where x_{ik} and q_{ik} represent respectively the quantity demanded and the quality of product variety produced by firm i for country k , and $\sigma > 1$ is the elasticity of substitution between varieties. The maximization problem results in a demand function for product-quality pair ik of the following form:

$$x_{ik}[Y_k, P_k, q_{ik}, po_{ik}] = \frac{Y_k}{P_k} (q_{ik}^{d_k})^{\sigma-1} \left(\frac{po_{ik}}{P_k} \right)^{-\sigma} \quad (2)$$

where Y_k is aggregate income and P_k is the quality-adjusted price index¹ in each country, which are both exogenous to the firm producing variety i under monopolistic competition, and po_{ik} is the final price of the product-quality pair. As such, demand is increasing in quality – in proportion to the quality preference parameter – and decreasing in price.

In the model of Kugler & Verhoogen (2012) inputs are assumed to be sourced exclusively from the local market. This is a clear simplification to maintain tractability, but as demonstrated by Demir (2013), similar results can ensue in a fully open economy set-up, although at the cost of losing the closed-form solutions. The main interest of our empirical analysis is to consider the impact on export performance of firms' sourcing behaviour based on imported intermediates, in particular related to the quality level of such imported intermediates. We continue to use the Kugler & Verhoogen model to motivate the empirical exercise since it provides elegant, closed form solutions that allow us to specify our hypotheses clearly, while also translating into an open economy setting, as shown by Demir (2013). In our empirical analysis, we further control for domestic intermediate purchases to ensure the robustness of our results to domestic sourcing. As in Kugler & Verhoogen (2012), there is an intermediate-input sector that transforms l labor-hours into heterogeneous intermediates of quality c . The cost of production directly depends on the quality of the intermediate good, with the production function in this sector being simply $F_l[l, c] = l/c$. We assume perfect competition in the market for intermediates and normalized homogeneous labor costs in all markets. The input price will therefore equal the marginal cost of production of intermediate suppliers, $p_l[c] = c$.

Each producer of final goods receives a capability draw $\lambda_i > 1$, which will affect both its production costs and its ability to produce quality. There is a fixed cost to produce for each destination, f_k , and the usual iceberg cost associated with shipping the product, τ_k . Production of physical units of the final good is given by $F_i[n] = n\lambda_i^a$, where n is the number of inputs used and $a > 0$ is the parameter describing how capabilities affect production efficiency. The marginal cost of a product serving each destination is therefore equal to $\frac{c_{ik} \tau_k}{\lambda_i^a}$, where c_{ik} is the desired

level of input quality firm i chooses to supply to market k .

As in the original models of Bastos et al. (2018) and Kugler & Verhoogen (2012), production of quality, q_{ik} , in the final good sector is determined by a CES combination of firm capability and input quality in the following way:

$$q_{ik}[\lambda_i, c_{ik}] = (\mu (\lambda_i^b)^{\theta} + (1-\mu)(c_{ik}^2)^{\theta})^{\frac{1}{\theta}} \quad (3)$$

where $\theta < 0$ indicates the scope of complementarity between capability and input quality, b indicates the scope for quality differentiation and μ is a multiplicative weight between the two inputs. The more negative θ is, the higher are complementarities between the two inputs. The scope for quality differentiation (b) is assumed to be positive and can be thought as sector-specific – a possibility we explore later but simplify away here by assuming it to be the same for all firms. We interpret the quality function (3) as a simplified product innovation production process by which firms upgrade and adapt their products for each market.

Firms will choose prices and the quality of its inputs in order to maximize profit in each production line according to the following equation:

$$\pi_{ik}[\lambda_i, c_{ik}, po_{ik}] = \left(po_{ik} - \frac{c_{ik} \tau_k}{\lambda_i^a} \right) x_{ik} - f_k \quad (4)$$

where we assume real wages are the same in all countries, ignoring exchange rates.² This set-up results in the following solutions for input and output quality, final prices and revenues for each product-quality (firm-country) pair:

$$c_{ik} = \left(\frac{(1-\mu)}{\mu} (2d_k - 1) \right)^{\frac{1}{\theta}} \lambda_i^{\frac{b}{\theta}} \quad (5)$$

$$q_{ik} = \left(\frac{1}{\mu} \frac{(2d_k - 1)}{2d_k} \right)^{\frac{1}{\theta}} \lambda_i^{\frac{b}{\theta}} \quad (6)$$

$$po_{ik} = \left(\frac{\sigma}{\sigma-1} \right) \tau_k \left(\frac{(1-\mu)}{\mu} (2d_k - 1) \right)^{\frac{1}{\theta}} \lambda_i^{\left(\frac{b}{\theta} - a \right)} \quad (7)$$

$$r_{ik} = \Gamma_1 \Gamma_2 \Gamma_3 \lambda_i^{\Gamma_4} \quad (8)$$

where

$$\Gamma_1 = \frac{Y}{P^{(1-\sigma)}} \left[\left(\frac{\sigma}{\sigma-1} \right) \tau_k \right]^{-(\sigma-1)} \left[\frac{\mu}{(1-\mu)} \right]^{\frac{-(\sigma-1)}{2\theta}},$$

$$\Gamma_2 = \frac{1 - d_k(\sigma-1)}{\mu},$$

$$\Gamma_3 = (2d_k)^{\frac{d_k(\sigma-1)}{\theta}} (2d_k - 1)^{\frac{-(2d_k-1)(\sigma-1)}{2\theta}} \text{ and,}$$

$$\Gamma_4 = \left[\frac{(\sigma-1)}{2} \right] [b(2d_k - 1) + 2a].$$

Once we assume $d_k > \frac{1}{2}$ to obtain an interior solution, it is straightforward to observe in Eqs. (5)–(8) that all variables are positive functions of firm capability, except for the price of final goods, which will be positive if $\frac{b}{2} > a$ in Eq. (7). These closed-form solutions also allow us to derive propositions relating firm capability to their selling and sourcing behavior.

¹ This is the usual price index adjusted by the quality of each variety, i.e. $P_k =$

$$\left(\int_{i \in \Omega} \left(\frac{P_{ik}}{q_{ik}} \right)^{1-\sigma} di \right)^{\frac{1}{1-\sigma}}.$$

² Production happens in the home country and is exported to $k-1$ destinations. Therefore, revenues have to be converted to the local currency, as in Bastos et al. (2018).

2.2. The effects of capability and the scope for quality differentiation

Eqs. (5) and (8) demonstrate that the quality of intermediates (c_{ik}) and firm revenue (r_{ik}) increase in firm capability ($\frac{\Delta c}{\Delta \lambda}, \frac{\Delta r}{\Delta \lambda} > 0$). Considering that c_{ik} summarizes the quality level of a bundle of inputs and, in practice, indicates a more intensive use of higher quality intermediates relative to lower quality ones, we can conclude that more capable firms will choose higher quality intermediates, sell higher quality products and have higher revenues.

Increasing the quality of products increases demand, but it also drives up production costs and counteracts these positive effects on sales. First-order conditions impose that firms will adjust the quality of their inputs until marginal gains in revenues are exhausted, i.e. $\frac{\Delta r}{\Delta c} = 0$, in equilibrium (for simplicity, we suppress subscripts from here on). However, it is still possible to demonstrate that more capable firms will tend to benefit proportionally more from increasing the quality of their inputs in terms of their revenues, i.e. that more capable firms not only increase the quality of their inputs but are also able to obtain proportionally more revenues while doing so. We can see that by exploring what happens to the ratio $\frac{r}{c}$ in equilibrium at different levels of firm

capability, i.e. solving the derivative $\frac{\Delta(\frac{r}{c})}{\Delta \lambda}$, which results in the following equation:

$$\frac{\Delta(\frac{r}{c})}{\Delta \lambda} = \Gamma_2 \Gamma_5 \Gamma_6 \Gamma_7 \lambda^{\Gamma_7-1} \quad (9)$$

where

$$\Gamma_5 = \frac{Y}{P^{(1-\sigma)}} \left[\left(\frac{\sigma}{\sigma-1} \right) \tau \right]^{-(\sigma-1)} \left[\frac{\mu}{(1-\mu)} \right]^{\frac{\sigma}{\sigma-1}},$$

$$\Gamma_6 = (2d)^{\frac{d_k(\sigma-1)}{\theta}} (2d-1)^{\frac{1-(2d_k-1)(\sigma-1)}{2\theta}}, \text{ and}$$

$$\Gamma_7 = \frac{1}{2} [b(2d-1)(\sigma-1) + (\sigma-1)2a - b].$$

While the terms Γ_5 and Γ_6 are unmistakably positive, Γ_7 is not necessarily positive and depends mainly on the values of d and σ , which is not surprising considering these parameters determine the opportunities for firms to increase revenue by increasing the quality of their products. It is the case, however, that for realistic values of the model parameters within the support we have defined, Γ_7 will tend to be positive.³ In essence, this result indicates that, as firm-capability increases, there will be a transition to a new equilibrium in which the impact of increasing the quality of intermediates on export revenues will also tend to be higher.

The reasoning above allows us to extract our first two hypotheses: (i) there will be a positive effect of firm capability on export revenues ($\frac{\Delta r}{\Delta \lambda} > 0$) and (ii) more capable firms will tend to benefit proportionally

more from using higher quality intermediates ($\frac{\Delta(\frac{r}{c})}{\Delta \lambda} > 0$). Moreover,

³ Γ_7 will only be negative if $d < \frac{1}{2} \left(\frac{\sigma}{\sigma-1} \right) - \frac{a}{b}$. Estimates of σ in the literature however, tend to indicate relatively high values for this parameter. Anderson & Van Wincoop (2004) review the literature and indicate [5,10] as a reasonable range, which is very close to the range usually assumed by empirical studies of product quality (Fan et al., 2015; Khandelwal et al., 2013; Manova & Yu, 2017). Considering the values of a , b and d are constrained to obtain internal solutions for the model ($a, b > 0$; $d > \frac{1}{2}$), our equations indicate either an inexistent or very strict range of values of d close to $\frac{1}{2}$ for which Γ_7 takes non-positive values.

Eq. (9) shows that the ratio $\frac{r}{c}$ is changing in firm capability (λ) in a non-linear manner, which also suggests that our choice of a threshold regression model as our preferred empirical strategy in the following sections is consistent with theory.⁴

This model also allows us to explore the effect of the scope for quality differentiation – represented in the theoretical model by the parameter b – on firms' sourcing and revenue results. Different sectors and products might offer different opportunities for quality improvements depending on the characteristics of their markets, such as the established level of standardization that firms must follow. This is further explored in the empirical section where we divide exports into customized and non-customized products to examine whether the relationship between imports and exports, and the mediating role for technological capabilities, differs between these different types of products. In the model, this idea

can be represented by the mixed partial derivative $\frac{\Delta^2(\frac{r}{c})}{\Delta \lambda \Delta b}$ depicted by Eq. (10), which indicates the influence of the scope for quality differentiation on the effects of firm capability:

$$\frac{\Delta^2(\frac{r}{c})}{\Delta \lambda \Delta b} = \frac{1}{2} \Gamma_2 \Gamma_5 \Gamma_6 \lambda^{\Gamma_7-1} [(2d-1)(\sigma-1) - 1] (1 + \Gamma_7 \ln \lambda) \quad (10)$$

The sign of (10) depends on the model parameters, but once again within realistic values of σ (see footnote 3) the expression above will tend to assume positive values. If that is the case, more capable firms will use higher quality inputs and produce higher quality products more intensively when there are more opportunities for quality differentiation. This idea is similar to previous results of the literature (Bas & Strauss-kahn, 2015; Fan et al., 2015) and provides us with an additional hypothesis, i.e. (iii) the influence of firm capability on the relationship between export revenues and the quality intermediates will be stronger for exports of products that have a higher scope for quality differentiation relative to products with fewer opportunities for differentiation. We explore our hypotheses empirically in the subsequent sections.

3. Empirical methodology

3.1. The linear model

Unlike most previous related work, we do not consider the firm capability to be completely unobserved and instead rely on a direct proxy for firms' technological and innovation skills. This variable, which we call *Tecper*, is constructed from information about the functions performed by employees in each firm, which are extracted from Brazilian administrative data and explained in greater detail in Section 4. This approach allows us to develop a simple framework that addresses the propositions we advance from the theoretical model simultaneously. These propositions summarize important aspects of the influence of innovation on firms' export performance and the role of complementarities between technological capabilities and imports.

Initially, we use a linear model for firms' export revenues that allows for both a direct effect of technological capabilities on export revenues and an indirect effect that depends on the presence of complementarities between capabilities and imported intermediates, the latter of which is captured by an interaction term between these variables. We estimate all models with time and firm fixed effects, further lagging all dependent variables to allow for delayed effects on exports and to reduce the simultaneity bias between exporting and importing. Despite this, we do not have a valid instrument for imports that would allow us to fully control for endogeneity in our model, and thus we remain cautious in interpreting our results in terms of correlations and not causal effects.

The firm-level time-varying controls include firm size (i.e. the

⁴ It is straightforward to show that $\left(\frac{\Delta^2(\frac{r}{c})}{\Delta \lambda^2} \neq 0 \right)$ under the support we defined for the parameters.

number of employees), the skill level of labour (i.e. average years of schooling), total factor productivity and capital-intensity. This set constitutes a reasonably complete group of variables that have consistently been found to positively correlate with export activity in the firm-level international trade literature (Bernard and Jensen, 1999, 2004; Greenaway and Kneller, 2007; Wagner, 2007). The effect of firm size is related to performance and to several other firm-level characteristics. Controlling for size is also necessary since our equation is estimated in levels and larger firms are more able to dilute the fixed-costs associated with entering into foreign markets, such as those related to distribution networks and marketing investments (Aristei et al., 2013; Bernard & Jensen, 2004; Pisu & Muûls, 2009; Roberts & Tybout, 1999). Average years of schooling controls for skill intensity and contrasts overall differences in the quality of labour to the presence of functions related to technology management by the firm, which are correlated but nevertheless different. The role of human capital is frequently captured in the literature through average wages, which are conditioned by the skill level of the labour force and usually found to positively affect firm export propensity and performance (Barrios et al., 2003; Dosi et al., 2015; Greenaway, Sousa, & Wakelin, 2004).

The inclusion of total factor productivity as a control variable is important due to its role in determining firm performance in foreign markets, strongly confirmed in both theoretical models and empirical investigations (Antràs & Helpman, 2004; Melitz, 2003; Wagner, 2007). More productive firms self-select into foreign markets and obtain higher revenues in both local and external markets, even after controlling for size (Hallak & Sivadasan, 2013). Total factor productivity is, of course, another proxy for capability, especially for capabilities related strongly to cost efficiency, whereas technical and scientific personnel indicate innovation capabilities and skills required to upgrade quality. Despite being necessary, this framework can be seen as creating a partition of capabilities of the sort formalized by Hallak & Sivadasan (2013), where two types of firm-specific capability are modelled: one influencing 'process productivity' and the other 'product productivity' or quality production. In the context of the Kugler & Verhoogen (2012) model, such a partition would potentially influence some theoretical predictions, especially those related to prices of final goods, a fact highlighted by the authors in their original version. However, both approaches converge in terms of the expected patterns relating capabilities to export revenues and input quality, and as such we consider this to be a minor concern for our analysis.⁵ We confirm this by running our different regression models excluding the total factor productivity variable, with results largely unaffected by its exclusion. The resulting estimation equation takes the following form:

$$\ln EX_{it} = \beta_0 + \beta_1 \ln IM_{it-1} + \beta_2 Tecper_{it-1} + \beta_3 \ln IM_{it-1} \times Tecper_{it-1} + \beta_4 \ln(K/L)_{it-1} + \beta_5 \ln EMP_{it-1} + \beta_6 SCHOOL_{it-1} + \beta_7 \ln TFP_{it-1} + \alpha_i + \nu_t + \mu_{it} \quad (11)$$

where $\ln EX$ is the log of exports,⁶ $\ln IM$ is the log of imported intermediates, $Tecper$ indicates technical and scientific personnel intensity and represents the proxy for firms technological capabilities, $\ln(K/L)$ is the log of the capital intensity, $\ln EMP$ the log of employees, $SCHOOL$ is the average years of schooling of workers, $\ln TFP$ the log of total factor productivity, and α_i and ν_t are firm and time-specific fixed-effects,

⁵ We could rationalize the effect of cost-related capabilities in the model by assuming them to be a firm-specific multiplicative parameter reducing cost, in which case it could enter the equations in a way similar to (the inverse of) the iceberg export cost. Our conclusions would not be affected as long as the new parameter is positive. Consequences, however, would be more relevant for export entry in a complete general equilibrium framework.

⁶ Note that we use $\ln(1 + EX_{it})$ and $\ln(1 + IM_{it})$ in our analysis to allow for the inclusion of firms with zero exports and imports.

respectively. Initially, we assume that imported intermediates are a proxy for higher quality intermediates, an approach based upon previous findings indicating that inputs sourced from abroad are of a higher quality when compared to local ones (Kugler and Verhoogen, 2008, 2009). Our subsequent step, explained in the following sections, is to account directly for the quality of these imports, and the results we obtain are compatible with the initial approximation done in this exercise.

3.2. Threshold effects: direct and indirect effects of firm capability

There is no reason to assume that the effect of capabilities on input quality increases linearly and, depending on their functional form, these non-linearities will not be captured properly by the interaction term. A more flexible way of modelling them is to consider that they represent classes of observations delimited by thresholds that depend on the level of firms' technological capabilities. Hansen (1999) developed a method to estimate unknown thresholds determined by the value of an observable variable in non-dynamic panels with individual-specific fixed effects. These thresholds emerge from the data and do not depend on specific assumptions regarding the functional form of non-linearities, which is a clear advantage compared to linear interactions. We describe our procedures to estimate thresholds effects in detail in Appendix II. We begin by allowing the parameter associated with imported intermediates to vary discretely depending on the intensity with which firms employ technical and scientific personnel ($Tecper$). We adjust this model initially with a single threshold, described by the following equation:

$$\ln EX_{it} = \beta_0 + \delta_1 \ln IM_{it-1} I(Tecper_{it-1} \leq \lambda_1) + \delta_2 \ln IM_{it-1} I(Tecper_{it-1} > \lambda_1) + \beta_1 Tecper_{it-1} + \beta_2 \ln(K/L)_{it-1} + \beta_3 \ln EMP_{it-1} + \beta_4 SCHOOL_{it-1} + \beta_5 \ln TFP_{it-1} + \alpha_i + \nu_t + \mu_{it} \quad (12)$$

where δ_1 and δ_2 are respectively the coefficients on $\ln IM$ in the low and high regimes, defined according to the cutoff value λ_1 of $Tecper$. The impact of importing intermediates is given by δ_1 for observations where $Tecper$ is not above λ_1 , which we call the low regime, while δ_2 describes the impact for observations in the high regime, where $Tecper$ is above λ_1 . We follow Hansen (1999) and estimate λ_1 by least squares, estimating Eq. (12) iteratively for values of $\hat{\lambda}_1$ above zero and below the 90th percentile of $Tecper$, and choosing the value of $\hat{\lambda}_1$ that minimizes the sum of squared errors. This method allows firms to appear in different regimes at different points in time, since the thresholds are determined at the observation (firm-year) level.

Once we obtain $\hat{\lambda}_1$, we must determine if it is statistically significant. If $\delta_1 = \delta_2$, the threshold is not significant, while the rejection of the null hypothesis indicates the presence of a non-linearity. However, since the threshold is not identified under the null hypothesis, the distribution of the test is non-standard. Once again, we follow Hansen (1999) and use a bootstrap procedure to obtain a p-value for the test and adopt the significance level of 5% to reject the null hypothesis of linearity. Once we obtain the first threshold, we start the search for a second threshold and the model takes the following form:

$$\ln EX_{it} = \delta_1 \ln IM_{it-1} I(Tecper_{it-1} \leq \lambda_1) + \delta_2 \ln IM_{it-1} I(\lambda_1 < Tecper_{it-1} \leq \lambda_2) + \delta_3 \ln IM_{it-1} I(Tecper_{it-1} > \lambda_2) + \beta X_{it-1} + \alpha_i + \nu_t + \mu_{it} \quad (13)$$

where X includes the same co-variables as in (12), δ_1 , δ_2 and δ_3 are the coefficients for $\ln IM$ in the three different regimes, defined according to the cutoff values, λ_1 and λ_2 , of $Tecper$. The impact of importing intermediates is now given by δ_2 for observations in the middle regime (i.e. between λ_1 and λ_2) and by δ_3 for observations in the high regime (i.e. above the λ_2 level of $Tecper$). A similar bootstrap procedure is followed to test if $\delta_3 = \delta_2$. We further re-estimate λ_1 since the initial estimate $\hat{\lambda}_1$ is not

efficient in the presence of a second threshold.

The differences in the estimates $\hat{\delta}_1$, $\hat{\delta}_2$ and $\hat{\delta}_3$ potentially capture the presence of complementarities between imported inputs and technological capabilities. When there are complementarities, we expect the effect of imported inputs to increase with capabilities ($\hat{\delta}_3 > \hat{\delta}_2 > \hat{\delta}_1$). However, the use of the threshold approach in the place of interaction terms means that we do not need to constrain the functional relationship to work in the same way (or to be significant) along the entire distribution of technological capabilities. We limit our search to two thresholds for the remainder of this work, which is enough to test our hypotheses. In the few cases in which we search for a third threshold we do not find the estimate to be significant.

3.3. Threshold effects: different quality levels of imports

The available empirical evidence indicates that imported inputs tend to be of higher quality in comparison to local ones, however, our dataset allows us to obtain better quality proxies for imported inputs. These measures supply us with an additional and more convincing test for the presence of complementarities between capabilities and input quality. Our main approach follows the bulk of the trade literature and assumes quality is an unobserved attribute of a product that incentivizes buyers to purchase higher quantities of a vertical variety irrespective of possibly higher prices charged by producers. Like most related works, we follow [Khandelwal et al. \(2013\)](#) and estimate an OLS equation for our dataset of imported products at the NCM 8-digit level,⁷ as depicted below, which can be obtained from a demand function similar to (2):⁸

$$\ln X_{ipkt} + \sigma \ln p_{ipkt} = \alpha_p + \alpha_{kt} + \varepsilon_{ipkt} \quad (14)$$

where X_{ipkt} accounts for the quantity and p_{ipkt} for the unit price imported by firm i of each 8-digit product p from country k in the year t . The product fixed effect α_p controls for differences in prices and quantities across different products, whereas the country-year fixed effects α_{kt} control for differences in income and price index across countries. The quality of each variety is obtained from the residual of (14), estimated as

$$\ln \hat{q}_{ipkt} = \frac{\varepsilon_{ipkt}}{\sigma - 1}.$$

Our objective is to expand the previous analysis by comparing complementarities between firm capability and imported inputs with different quality levels. In order to do so, we classify each imported variety as being of high (low) quality if its estimated quality is above (below) the median of imports,¹⁰ therefore creating two import variables at the firm-level: high-quality and low-quality imports. Sourcing more high-quality inputs should signal a clear increase in the quality of firms' inputs. We later check this approach using three alternatives: estimating quality controlling for product-year (α_{pt}) and country fixed effects (α_k); estimating quality using simple unit prices and once again classifying high (low) quality flows as those above (below) product-year unit price averages; and replacing imported input value by the weighted average of the quality index of imports. These latter approaches are

⁷ The NCM classification (Common Mercosur Taxonomy) is used by the Mercosur countries and classifies products in 8-digit codes, where the first 6 digits correspond to the Harmonized System.

⁸ Eq. (14) is standard and can be obtained from a demand function such as $x_{ipkt} = \frac{Y_k}{P_k} (q_{ipkt})^{\sigma-1} \left(\frac{p_{ipkt}}{P_k} \right)^{-\sigma}$, whose main differences with regard to (2) are allowing firms to produce more than one product variety and simplifying away the country-specific exponent d .

⁹ We show results for $\sigma = 5$, which is in line with the most popular empirical estimations ([Anderson & Van Wincoop, 2004](#)), but our results do not appear to be sensitive to different values within an empirically reasonable range.

¹⁰ The distribution of quality is not very skewed, therefore the average is quite close to the median. Moreover, our results are not affected by using the 40th or 60th percentiles or the use of the median of intermediates only.

explained further in the robustness section, but our results nevertheless remain consistent across.

With these two import variables, we must now estimate thresholds for two non-linear variables. We follow analogous stepwise procedures, which are also explained in [Appendix II](#), and estimate one threshold at a time. The model with the highest number of thresholds takes the following form:

$$\begin{aligned} \ln EX_{it} = & \delta_1 \ln IM_{HQ_{it-1}} I(Tecper_{it-1} \leq \lambda_1) + \delta_2 \ln IM_{HQ_{it-1}} I(\lambda_1 < Tecper_{it-1} \\ & \leq \lambda_2) + \delta_3 \ln IM_{HQ_{it-1}} I(Tecper_{it-1} > \lambda_2) + \delta_4 \ln IM_{LQ_{it-1}} I(Tecper_{it-1} \\ & \leq \lambda_3) + \delta_5 \ln IM_{LQ_{it-1}} I(\lambda_3 < Tecper_{it-1} \\ & \leq \lambda_4) + \delta_6 \ln IM_{LQ_{it-1}} I(Tecper_{it-1} > \lambda_4) + \beta X_{it-1} + \alpha_i + \nu_t + \mu_{it} \end{aligned} \quad (15)$$

where $\ln IM_{HQ}$ and $\ln IM_{LQ}$ are now the log of imported intermediates of high and low quality, respectively. Additionally, we estimate models (12), (13) and (15) considering as the dependent variable the revenue from the exports of customized exports and of non-customized products separately.¹¹ Following previous studies ([Bas & Strauss-Kahn, 2015](#); [Fan et al., 2015](#); [Flach, 2016](#)), we use the [Rauch \(1999\)](#) classification of products to differentiate exports between these two categories of goods. The former should present a wide scope for quality differentiation, while the competition in the latter would be mostly governed by cost factors, allowing lower opportunities for product innovation. This provides us with a test for prediction (iii) of the model developed in [Section 2](#).

4. Data

This work is based on data from three different sources: (i) the Brazilian Annual Industrial Survey (PIA); (ii) the Annual Registry of Social Information (RAIS); and (iii) the registry of foreign trade of the Secretary of Foreign Trade (Secex). PIA is the most representative database of Brazilian firms. It annually collects detailed business information (revenues, expenses, personnel, wages, investments, etc.) of Brazilian firms in manufacturing and extractive industries. All firms with more than 30 employees or revenues above R\$ 13.6 million in the previous year are surveyed, while firms below these thresholds are sampled. We exclude sampled firms and those from extractive sectors and concentrate only on firms from within manufacturing. This sample is merged with information about firm age, number of employees, average years of study and employee function taken from RAIS, which collects labour information from all active Brazilian firms.

Finally, the resulting sample is merged with the registry of foreign trade (Secex), containing administrative information for all exporting and importing firms (value, product, quantity and destination) from 1997-2007. We restrict our analysis to the 2001-2007 period which is the most recent period for which trade data is available. Most of the empirical analysis involves lagging the dependent variable, which results in a loss of some observations. Nevertheless, the resulting unbalanced panel contains 146,862 entries for 39,307 firms in total and approximately 60% of Brazilian accumulated exports during the period. All data was accessed at the Brazilian Institute of Geography and Statistics (IBGE) safe room and confidentiality procedures were followed. When estimating the quality of varieties, we lose observations for 2001 because unit prices are not available for that single year.

¹¹ We employ the conservative version of the [Rauch \(1999\)](#) classification of differentiated products, which has become quite popular in the literature as a measure of complexity or 'contract-intensity' of products ([Andersson & Weiss, 2012](#); [Antràs & Chor, 2013](#); [Del Prete & Rungi, 2015](#)). His methodology consists of dividing products in three categories: traded in organized exchanges, reference priced in trade publications, and all others. The first two categories indicate homogeneous products traded in dense markets (non-customized), while the residual identifies differentiated products more likely to be traded in networks (customized).

The value of exports and imports were converted to Brazilian currency using average annual exchange rates and all nominal variables were deflated using 3-digit sector level price indexes (IPA-OG) supplied by the Getulio Vargas Foundation (FGV). The capital stock is estimated using the perpetual inventory method. Total factor productivity (TFP) is estimated according to the Levinsohn & Petrin (2003). This method, as well as other semiparametric approaches routinely used in the literature (Akerberg et al., 2006; Olley & Pakes, 1996), corrects the bias caused by the endogeneity between variable inputs and unobserved firm productivity. Our TFP estimations are the same as used in Araújo (2016) and Messa (2014).

In Appendix I, Table I.1, we depict the number, average exported value and average number of country-product pairs exported and imported by Brazilian firms each year. The percentage of traders is high relative to the overall population of Brazilian firms (Kannebley Jr et al., 2009), with around 26% (23%) of firms exporting (importing). This is due to our sample being limited to firms with more than 30 employees or revenues above R\$ 13.6 million, therefore excluding very small firms. The number of exporters and importers tends to grow, albeit with quite some variation across years. These movements might be related to exchange rate variations, which are rather sharp in Brazil. Nevertheless, we control for these effects using time fixed effects in all regressions.

One of the major differences from our work in comparison to previous contributions is to proxy firms' technological capabilities directly. We explore the richness of our dataset to identify specific *occupations* of employees within the firm that are likely related to advanced technical and scientific activities. These include employees classified as researchers, engineers, R&D managers, IT personnel, physicists, chemists and other science-related functions.¹² This variable was developed by Araújo et al (2009), who showed that it is highly correlated with firms' internal R&D expenditures as declared in innovation surveys.¹³ We divide the number of technical and scientific personnel by total firm employment to obtain an intensity variable, i.e. the *Tecper* variable described above.

This approach is similar in nature to the one used by Harrigan et al (2016) to proxy the adoption of new technologies, although we include a broader range of occupations compared to the ICT-related "techies" they concentrate on. Compared to usual measurements of labour quality, this type of approach has the advantage of separating science and engineering personnel from qualified overall white-collar staff, providing a better proxy for internal functions associated with innovation activities. The use of *occupations* helps to separate the qualification level of employees from the activities effectively performed by them in the firm. Our objective is to proxy the presence of 'intermediate' and 'advanced' capabilities related to the management of technological change (Lall, 1992), which bear a close link to innovation activities in the firm. Therefore, we use this variable as an indicator of innovation activities, which despite the richness of our dataset, we cannot observe directly at the firm-level.

As with any proxy for capabilities, there are limitations to our approach. For instance, although we purposefully include a wide array of occupations in the *Tecper* variable, it is likely that the set we choose does not reflect the necessary skills and innovation patterns of many sectors. Indeed, only 29% of observations have positive values for technical and scientific personnel in our sample. Simultaneously, *Tecper*

will reflect innovation partially if only more radical changes involve an expansion in technical and scientific personnel and firms use less qualified employees for more incremental or routine changes, or for instance hire outside consultants. While we cannot rule these problems out entirely, we do provide results replacing *Tecper* by the intensity of university degree personnel in the firm in Table 8, which is a more general variable and present in the vast majority of firms, albeit still highly correlated to *Tecper*. The results we find are highly compatible, although somewhat weaker compared to our preferred variable.

The average intensity of technical and scientific personnel is 0.7% of total employees for the overall population, with an average intensity of 2.3% for firms that have any positive values of *Tecper*. Appendix I, Table I.2, presents descriptive statistics for the main variables used in the empirical section, differentiating between firms that have technical and scientific personnel and those that do not. In Fig. 1, we compare the density of productivity, exports and imports of intermediates between firms that have positive and zero values for technical personnel. As expected, the density distribution of firms that have technical personnel clearly dominates the remaining firms in the case of all three firm variables, reinforcing previous indications that technological capabilities are associated with a higher performance both in local and international markets.

5. Results

5.1. The linear model

In Column (1) of Table 1, we present results for the complete linear model of Eq. (11). Control variables are all positive and significant. Increasing the number of employees by 1% is associated with a 0.65% increase in exports, whereas the same percentage increase in total factor productivity and capital intensity is associated with a 0.25% and 0.012% increase in exports, respectively. A one-year increase in average schooling of employees has a sizeable effect, being associated with an increase in foreign sales of 7.1%. Regarding our variables of interest, a 10% increase in imported intermediates is associated with a 0.53% increase in exports, which although not large is a significant effect. The *Tecper* variable presents a strong and significant direct correlation: a 1 percentage-point increase in *Tecper* is associated with a 2.86% increase in exports. The interaction term capturing complementarities between imports and innovation is positive but remains non-significant in all estimations, indicating either the absence of complementarities or that these are not being properly estimated by the linear interaction term.

In Column (2) we exclude the TFP variable since this could be considered an alternative form of firm capability as well as generally being a performance variable similar to export revenue. When doing this, the coefficients on most variables increase, especially that on employment (size). This signals the influence of efficiency and cost selection in the model, as expected due to the self-selection of more productive firms in export markets. The coefficient on the interaction term increases in size by a factor of three but remains insignificant. In Column (3) we also include the intensity of personnel with a college degree in the firm to test to what extent *Tecper* indeed represents firms' technological capabilities or just proxies for white-collar workers in the firm. As expected, the two variables most correlated with college personnel – *Tecper* and *SCHOOL* – have their coefficients reduced when compared to Column (1), although they remain highly significant. Moreover, the effect of *Tecper* is almost two times higher than that of college degree personnel, which increases our confidence in *Tecper* as a useful proxy for capability.

As we saw in Section 2, we have good reasons to assume that complementarities between technological capabilities and input quality have a more complex relationship, likely dependent on the level of technological capabilities itself. In this case, the linear interaction term is not well equipped to test this hypothesis. The first hint in that direction is given in Column (4), where we replace *Tecper* by a binary version

¹² The *Tecper* variable is constructed based on the Brazilian Classification of Occupations (CBO). It includes the following functions (CBO codes between parenthesis): researchers (203), engineers (202, 214, 222), R&D directors (1,237) and R&D managers (1,426), biotechnologists, geneticists, meteorologists (201), mathematicians, statisticians and similar (211), IT professionals (212), physicists, chemists and similar (213), biologists and similar (221). The most numerous sub-group is engineers.

¹³ This correlation varied between 0.83 and 0.92, depending on the wave of the innovation survey used.

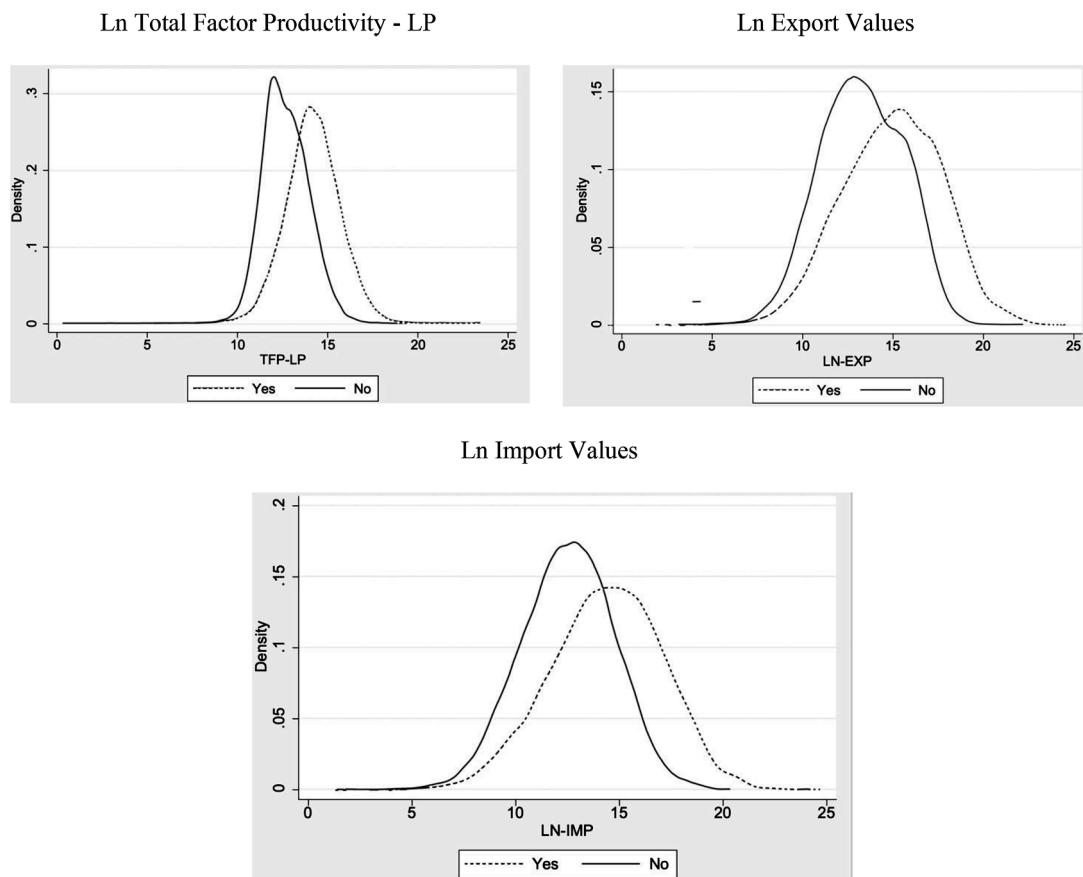


Fig. 1. Kernel density estimates for firms with and without technical personnel (*Tecper*).

Notes: Kernel densities for firms that have technical and scientific personnel (dotted line) and firms that do not (unbroken line). The dotted curve dominates the unbroken curve along the entire distribution. Only firms with non-zero export (import) values are included in the figure for the export (import) density, which is not the case for summary statistics in Table I.2. We still add one to exports and imports in the log transformation.

that takes the value one for any positive value of technical and scientific personnel in the firm. Results suggest that the correlation between imported intermediates and exports is indeed higher for firms that employ technical and scientific personnel, although the dummy variable itself is non-significant. This serves as a further motivation to look deeper into these non-linearities using less strict assumptions about their functional form.

5.2. Threshold effects for imported intermediates

Column (1) in Table 2 depicts results from a linear FE-OLS model without thresholds for comparison, while in Column (2) results for the one threshold case of Eq. (12) are shown. This threshold is found very close to the zero cutoff, at 0.06% of *Tecper*. The bootstrapping procedure indicates a p-value of 0.00, which allows us to reject the hypothesis of linearity. This implies that very low levels of technical personnel already change the effect of imported intermediates from 0.044 to 0.064 on average – a 45% percent difference between the low regime and the high regime. Since this threshold is very close to zero, it is almost equivalent to the dummy interaction model that we estimated in Column (6) of Table 1. Indeed, the marginal effect of imported intermediates is almost the same in the two models.¹⁴

This result confirms the presence of complementarities, but we

would like to know if they continue to increase as the level of technological capabilities grow or if any positive level of *Tecper* allows firms to reap more benefits from importing. As such, we estimate Eq. (13), with the results reported in Column (3) of Table 2. The second threshold found at a value of *Tecper* of 1.17% is significant with a p-value of 0.001. An increase of 10% in imported intermediates is correlated with a future increase of exports of 0.72% in the high regime of *Tecper*, 0.58% in the medium regime and 0.44% in the low regime. The remaining co-variables remain largely unaffected in the regressions, with the exception of *Tecper*, whose impact is significantly reduced from 3% to around 2%, indicating that some of the effect of firms' technological capabilities on exports is indeed related to variations in imported intermediates.

These results suggest that complementarities continue to exist as firms accumulate more technological capabilities. We check if this process continues at even higher levels of *Tecper* by searching for additional thresholds, but no significant evidence of a further threshold was found. Column (4) shows the results for the estimation of a third threshold with four regimes of imports of intermediates. This extra threshold is found at 1.70% of *Tecper*, but the bootstrap procedure indicates a p-value of 0.11, allowing us to reject the hypothesis of a third non-linearity in the model. Therefore, model (3) remains our preferred formulation.

5.3. Threshold effects and the quality of imports

Results in Table 2 confirm both the existence of significant impacts of technological capabilities on firms' export performance and the presence of complementarities between imported intermediates and

¹⁴ The marginal effect of importing for firms that have positive technical and scientific personnel is 0.063 (0.044 + 0.019) in Table 1, column 4, compared to 0.064 in Table 2 column 2. The effect is the same (0.044) for firms that do not have technical and scientific personnel.

Table 1
Linear model (FE-OLS regressions).

VARIABLES	(1) Ln Exports	(2) Ln Exports	(3) Ln Exports	(4) Ln Exports
ln IM (-1)	0.053 (0.003)***	0.056 (0.003)***	0.053 (0.003)***	0.044 (0.003)***
Tecper (-1)	2.861 (1.028)***	2.766 (1.022)***	2.170 (1.040)**	
ln IM Tecper (-1)	0.016 (0.082)***	0.047 (0.081)***	0.020 (0.082)***	
ln EMP (-1)	0.654 (0.030)***	0.709 (0.029)***	0.665 (0.030)***	0.629 (0.029)***
ln K/L (-1)	0.012 (0.005)**	0.005 (0.005)	0.012 (0.005)**	0.012 (0.0049)**
SCHOOL (-1)	0.071 (0.014)***	0.076 (0.015)***	0.047 (0.015)***	0.073 (0.014)***
ln TFP (-1)	0.256 (0.019)***		0.256 (0.019)***	0.256 (0.018)***
College Personnel (-1)			1.3681 (0.322)***	
Tecper Dummy (-1)				0.060 (0.040)
Tecper Dm×ln IM (-1)				0.019 (0.004)***
Observations	146,862	146,862	146,862	146,862
Firm fixed-effects	YES	YES	YES	YES
Time fixed-effects	YES	YES	YES	YES
Number of groups	39,307	39,307	39,307	39,307
F-statistic	131.4	126.0	122.7	133.0
R-squared	0.39	0.36	0.39	0.39

*** p<0.01, ** p<0.05, * p<0.1. Standard-errors between parentheses. We add one to Exports and Imports for the log transformation. All models estimated with time and firm-specific fixed effects.

technological capabilities. These complementarities create nonlinearities that are not well described by linear interaction terms. An increase in technological capabilities from a low initial base increases the correlation between imported intermediates and export performance, with firms becoming more capable of managing technology. At the highest capability levels, however, the additional impact ceases to be relevant, indicating decreasing returns in the complementarities between capabilities and imported intermediates.

If exporters with higher technological capabilities benefit more from importing because of the superior quality and embodied technology of these inputs – which allows them to innovate and improve their products – we may expect that imports of a higher quality/technological content will present stronger complementarities with technological capabilities. To test for this hypothesis, we proceed to separate imported inputs according to the estimated quality level of these flows, as explained in Section 3. These results are depicted in Table 3. Column (1) shows results for a linear model,¹⁵ while columns (2) and (3) report results for the one and two threshold model for high-quality imports, respectively. Without accounting for thresholds, firms appear on average to benefit equally from importing high and low-quality intermediates in terms of their export performance. The remaining columns, however, show there are significant heterogeneities for firms with different levels of technological capabilities.

In column (2) the threshold for high-quality imports is estimated at 0.05% of *Tecper*, which is very close to the threshold for overall imports (found at 0.06%) and confirms that even firms at initial levels of the technological capability benefit more from imported intermediates. A one percent increase in the value of high-quality imported intermediates is associated with an increase in exports more than 2.6 times higher for

¹⁵ One should observe that both coefficients are below the value we obtain for overall imports in Eq. (1). Although the samples are different, most of the difference is due to the log transformation we use: the sum of the logs of high- and low-quality imports equals the log of their product and not the log of their sum. Therefore, these two models are not directly comparable.

Table 2
Threshold regressions for imported intermediates.

VARIABLES	(1) Ln Exports	(2) Ln Exports	(3) Ln Exports	(4) Ln Exports
ln EMP (-1)	0.654 (0.029)***	0.641 (0.029)***	0.642 (0.029)***	0.642 (0.029)***
ln K/L (-1)	0.012 (0.0043)**	0.012 (0.0046)**	0.012 (0.0046)**	0.012 (0.0046)**
SCHOOL (-1)	0.071 (0.014)***	0.070 (0.014)***	0.070 (0.014)***	0.070 (0.014)***
ln TFP (-1)	0.256 (0.018)***	0.256 (0.018)***	0.255 (0.018)***	0.255 (0.018)***
Tecper (-1)	3.015 (0.686)***	2.414 (0.695)***	1.986 (0.706)***	1.814 (0.7149)***
ln IM (-1)				
No Threshold	0.053 (0.003)***			
(Tecper ≤ 0.06%)		0.044 (0.003)***	0.043 (0.003)***	0.043 (0.003)***
(Tecper > 0.06%)		0.064 (0.003)***		
(0.06% < Tecper ≤ 1.17%)			0.058 (0.004)***	0.058 (0.004)***
(Tecper > 1.17%)			0.072 (0.004)***	
(1.17% < Tecper ≤ 1.70%)				0.068 (0.005)***
(Tecper > 1.70%)				0.075 (0.005)***
p-value (threshold)	-	0	0	0.105
Observations	146,862	146,862	146,862	146,862
Firm fixed-effects	YES	YES	YES	YES
Time fixed-effects	YES	YES	YES	YES
F-statistic	143.3	133.9	124.5	115.8
Within R-squared	0.014	0.015	0.015	0.015
R-squared	0.38	0.38	0.39	0.39

*** p<0.01, ** p<0.05, * p<0.1. Standard-errors between parentheses. We add one to Exports and Imports for the log transformation. All models estimated with time and firm-specific fixed effects.

firms that have some technical and scientific personnel compared to firms that have very low levels or none. In column (3) we search for an additional threshold, which is found when firms achieve 1.18% intensity of *Tecper*. At the highest level of technological capabilities, the correlation between high-quality imported intermediates and next period exports is therefore higher than in the medium and low regimes of capability by an order of 1.46 and 3.3, respectively.

The logic is precisely reversed with regard to low-quality imports. In Column (4) we report results when searching for a threshold for low-quality imports while maintaining the previous thresholds for high-quality imports. The estimated cut-off is found at a value for *Tecper* of 0.54% and is significant at the 5% level. At the high regime of capability, the estimate for the effect of low-quality imports on exports (0.024) is significantly smaller compared to the high regime of *Tecper* (0.038). We attempt to estimate a second threshold for this variable, but the model returns a p-value of 11% (not depicted). Therefore, model (4) remains our preferred equation.

It is therefore clear that the combination of high capabilities and high-quality imports shows a strong relationship with export revenues. As capability increases, low-quality intermediates have their effect slightly reduced, while overall imports increase their effect accompanying the higher coefficients for high-quality imports. These results therefore demonstrate relevant heterogeneities among firms and offer additional support for the idea that important complementarities exist between capability and the quality of the intermediates imported by local firms.

Table 3

Threshold regressions for imported intermediates of different estimated quality.

VARIABLES	(1) Ln Exports	(2) Ln Exports	(3) Ln Exports	(4) Ln Exports
ln EMP (-1)	0.576 (0.047)***	0.559 (0.046)***	0.560 (0.046)***	0.562 (0.046)***
ln K/L (-1)	0.018 (0.005)***	0.018 (0.005)***	0.018 (0.005)***	0.018 (0.005)***
SCHOOL (-1)	0.053 (0.018)***	0.053 (0.018)***	0.053 (0.018)***	0.053 (0.018)***
ln TFP (-1)	0.215 (0.025)***	0.215 (0.025)***	0.215 (0.025)***	0.215 (0.025)***
Tecper (-1)	3.484 (0.874)***	2.783 (0.849)***	2.322 (0.840)***	2.523 (0.843)***
ln IM_HQ (-1)				
No Threshold	0.033 (0.005)***			
(Tecper $\leq$ 0.05%)		0.018 (0.006)***	0.018 (0.006)***	0.015 (0.006)**
(Tecper > 0.05%)		0.049 (0.006)***		
(0.05% < Tecper $\leq$ 1.18%)			0.040	0.042
(Tecper > 1.18%)			0.007*** 0.059 (0.007)***	0.007*** 0.064 (0.007)***
ln IM_LQ (-1)				
No Threshold	0.033 (0.005)***	0.033 (0.005)***	0.033 (0.005)***	
(Tecper $\leq$ 0.54%)				0.038 (0.005)***
(Tecper > 0.54%)				0.024 (0.006)***
p-value (threshold)	-	0	0	0.01
Observations	125,176	125,176	125,176	125,176
Firm fixed-effects	YES	YES	YES	YES
Time fixed-effects	YES	YES	YES	YES
F-statistic	45.81	43.55	40.54	38.13
R-squared	0.40	0.40	0.41	0.41

*** p<0.01, ** p<0.05, * p<0.1. Standard-errors between parentheses. We add one to Exports and Imports for the log transformation. All models estimated with time and firm-specific fixed effects.

5.4. Different levels of customization in exported products

In Table 4 we turn to the case of different customization levels of exports. Customized products offer a higher scope for quality differentiation, which according to the model presented in Section 2 should intensify the influence of capabilities on the relationship between high-quality intermediates and revenues, compared to non-customized exports. In columns (1) and (2) we estimate the model for customized products with one and two thresholds and observe that complementarities with imported intermediates are present for these markets, as the effect of imported intermediates decreases from 0.065 to 0.038 once we move to lower capability firms in Column (2). In columns (3) and (4), we observe that there is also an increase in the effect of intermediates for non-customized products as firm capabilities increase, but the coefficients remain largely below those in columns (1) and (2) for similar regimes of technological capability, with the absolute difference increasing as we move to higher regimes of technological capabilities. Regarding the direct effect of technological capabilities, we observe that the coefficient for *Tecper* is always positive, but only significant (at the 5% level) in Columns (1) and (2), falling slightly in magnitude in Column (2). However, for exports of non-customized products, estimates for technological capabilities are insignificant and negative, suggesting that the direct effects of capabilities are irrelevant for non-customized exports.

This evidence so far confirms our expectations, but in Table 5 we go further and repeat the procedure of estimating thresholds on imports of different estimated quality in the case of customized and non-

Table 4

Threshold regressions for exports of customized and non-customized intermediates.

VARIABLES	(1) Ln Cust	(2) Ln Cust	(3) Ln Non Cust	(4) Ln Non Cust
Tecper (-1)	1.983 (0.775)**	1.818 (0.769)**	-0.028 (0.568)	-0.182 (0.571)
ln IM (-1)				
(Tecper $\leq$ 0.02%)		0.038 (0.005)***		
(0.02% < Tecper $\leq$ 1.28%)		0.051 (0.006)***		
(Tecper > 1.28%)	0.062 (0.006)***	0.065 (0.007)***		
(Tecper $\leq$ 1.28%)	0.043 (0.005)***			
(Tecper $\leq$ 0.14%)			0.011 (0.003)***	0.008 (0.003)***
(Tecper > 0.14%)			0.026 (0.005)***	
(0.14% < Tecper $\leq$ 1.42%)				0.018
(Tecper > 1.42%)				0.005*** 0.029 (0.005)***
p-value (threshold)	0	.01	0	0
Observations	146,862	146,862	146,862	146,862
Firm fixed-effects	YES	YES	YES	YES
Time fixed-effects	YES	YES	YES	YES
F-statistic	51.53	47.68	29.27	27.11
R-squared	0.31	0.31	0.25	0.25

*** p<0.01, ** p<0.05, * p<0.1. Standard-errors between parentheses. We add one to Exports and Imports for the log transformation. Control variables omitted are the same of Table 2.

customized exports. Although the direct effect of technological capabilities presents similar coefficients in all equations, the results for imported intermediates add important insights. Columns (1) and (2) depict results for one and two thresholds of low-quality imports already accounting for two thresholds of high-quality imports, which are confirmed to be significant. The hypothesis of a second non-linearity for low-quality imports is rejected at the 5% level (p-value 9%). Consistent with the results in Table 3, we find for low quality imports that the relationship with export revenues declines with increasing firm capabilities. The results further show strong complementarities between high-quality imports and capabilities, however, with estimates jumping from 0.013 to 0.06 between the low and high regimes of technological capabilities in column (1).

In columns (3) and (4), we depict results for non-customized exports. In both cases, we only show one threshold for high-quality imports, as we do not find a second significant threshold, which already indicates a lower degree of complementarities for these exports. Nevertheless, in column (4) we also estimate a second threshold for high-quality imports but once again the threshold is not significant (p-value of 6% in the bootstrap procedure), making the model in column (3) our preferred estimation in this case. Comparing this model with column (1) confirms that complementarities between imported inputs and firm capability are highly driven by high-quality imports and concentrated in exports of customized products. In the case of non-customized exports, complementarities are weak and not very different between high and low-quality imports. These results are strongly in line with the hypothesis (iii) of Section 2, support the presence of stronger complementarities between firm capability and the quality of the inputs they source for products with a higher scope for quality differentiation.

Table 5

Threshold regressions for exports of customized and non-customized intermediates, imported intermediates of different quality.

VARIABLES	(1) Ln Cust	(2) Ln Cust	(3) Ln Non Cust	(4) Ln Non Cust
Tecper (-1)	2.216 (0.760)***	1.994 (0.769)***	-0.171 (0.661)	-0.224 (0.662)
ln IM_HQ (-1) (Tecper $\leq$ 0.06%)	0.013 (0.004)***	0.014 (0.004)***		
(0.06% < Tecper $\leq$ 1.18%)	0.035 (0.004)***	0.036 (0.004)***		
(Tecper > 1.18%)	0.060 (0.005)***	0.058 (0.005)***		
(Tecper $\leq$ 1.35%)			0.013 (0.004)***	0.014 (0.004)***
(Tecper > 1.35%)			0.023 (0.006)***	0.018 (0.008)***
ln IM_LQ (-1) (Tecper $\leq$ 0.54%)	0.033 (0.003)***	0.032 (0.004)***		
(Tecper > 0.54%)	0.016 (0.004)***			
(0.54% < Tecper $\leq$ 1.83%)		0.013 (0.004)***		
(Tecper > 1.83%)		0.023 (0.006)***		
(Tecper $\leq$ 0.17%)			0.006 (0.005)***	0.005 (0.005)***
(Tecper > 0.17%)			0.017 (0.005)***	
(0.17% < Tecper $\leq$ 1.35%)				0.014 (0.006)***
(Tecper > 1.35%)				0.024 (0.008)***
p-value (threshold)	0	.09	0	.06
Observations	125,176	125,176	125,176	125,176
Firm fixed-effects	YES	YES	YES	YES
Time fixed-effects	YES	YES	YES	YES
F-statistic	54.61	51.19	37.35	34.91
R-squared	0.32	0.32	0.26	0.26

*** p<0.01, ** p<0.05, * p<0.1. Standard-errors between parentheses. We add one to Exports and Imports for the log transformation. Control variables omitted are the same of Table 2.

5.5. Destination markets and export quality

In Table 6 we perform two additional tests for the presence of complementarities between capabilities and the quality of imports. First, in columns (1) and (2), we divide exports in high- and low-quality, adapting directly the procedure used to estimate the quality of imports for the case of exports. This exercise is meant to explore more directly previous evidence showing that firms use higher quality inputs to upgrade the quality of their own products (Bas & Strauss-kahn, 2015; Manova & Zhang, 2012). In that sense, we expect to find higher estimates and stronger complementarities for high-quality imported inputs when high-quality exports are the dependent variable, similar to the results we found for customized intermediates.

In Column (1), we depict regressions for the preferred model for high-quality exports, which has one threshold for low-quality imports – the hypothesis of a second non-linearity is rejected for this variable – and two for high-quality imports. Column (2) reports the preferred model in the case of low-quality exports, which has no thresholds for low-quality imports and two thresholds for high-quality imports. The comparison between these columns confirms our expectations, as we can observe that the correlation between high-quality intermediates and high-

Table 6

Threshold regressions for exports according to destination and imports of customized and non-customized intermediates.

VARIABLES	(1) Ln Exp_HQ	(2) Ln Exp_LQ	(3) Ln Exp_OECD	(4) Ln Exp_NOECD
Tecper (-1)	2.012 (0.821)***	1.58 (0.777)**	1.706 (0.665)***	0.912 (0.700)
ln IM_HQ (-1) (Tecper $\leq$ π 1)	0.02 (0.004)***	0.011 (0.004)***		
(π 1 < Tecper $\leq$ π 2)	0.047 (0.005)***	0.031 (0.004)***		
(Tecper > π 2)	0.07 (0.005)***	0.048 (0.005)***		
ln IM_LQ (-1) No Threshold		0.017 (0.003)***		
(Tecper $\leq$ 0.48%)	0.04 (0.004)***			
(Tecper > 0.48%)	0.023 (0.005)***			
ln IM_OECD (-1) (Tecper $\leq$ λ 1)			0.019 (0.004)***	0.022 (0.004)***
(λ 1 < Tecper $\leq$ λ 2)			0.03 (0.005)***	0.039 (0.004)***
(Tecper > λ 2)			0.049 (0.004)***	0.052 (0.004)***
ln IM_NOECD (-1) No Threshold				0.032 (0.003)***
(Tecper $\leq$ 1.88%)			0.023 (0.003)***	
(Tecper > 1.88%)			0.012 (0.004)***	
p-value (threshold)	0	0	0	0.01
Observations	125,176	125,176	146,862	146,862
Firm fixed-effects	YES	YES	YES	YES
Time fixed-effects	YES	YES	YES	YES
F-statistic	66.63	50.19	116.2	63.30
R-squared	0.40	0.35	0.38	0.30

*** p<0.01, ** p<0.05, * p<0.1. Standard-errors between parentheses. We add one to Exports and Imports for the log transformation. Control variables omitted are the same of Table 2.

in column (1): π 1=0.06% and π 2 = 1.18%

in column (2): π 1=0.06% and π 2 = 1.91%

in column (3): λ 1=0.07%, λ 2=1.18%

in column (4): λ 1=0.09%, λ 2=1.30%

quality exports is higher in all regimes than the corresponding coefficients in the case of low-quality exports, with these differences being especially large for firms with higher capabilities. Consistent with earlier evidence, high capability firms appear to benefit less from using lower quality imports when considering the export of high-quality products, with no significant differences in the effect of low-quality imports driven by firm capability observed in the case of low-quality exports.

In columns (3) and (4) we replace the estimated quality of exports and imports by their origin, dividing trade flows from OECD and non-OECD countries¹⁶ The empirical evidence indicates that firms tend to

¹⁶ We exclude countries that entered the OECD during the 1990s and after, therefore classifying as OECD only the older and mostly wealthier members of the organization. These are Austria, Belgium, Canada, Denmark, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, The Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, United Kingdom, United States, Japan, Australia and New Zealand.

Table 7

Extension regressions, threshold regressions for exported value and country-product varieties of customized and non-customized intermediates, imported intermediates and capital goods.

DEP VARIABLES	(1) Ln Prod-Country Exp.	(2) Ln Cust Prod-Country Exp.	(3) Ln Non Cust Prod-Country Exp.	(4) Ln Exports	(5) Ln Prod-Country Imports	(6) Ln Exports	(7) Ln Exports
IMPORT VARIABLE	Ln Imports	Ln Imports	Ln Imports	Ln Prod-Country Imports	Ln Prod-Country Imports	Ln Imports	Ln Imports
Tecper (-1)	0.356 (0.109)***	0.388 (0.144)***	-0.08786 (0.088)	2.144 (0.0704)***	0.351 (0.109)***	2.565 (0.695)***	2.521 (0.695)***
ln IM (-1)							
(Tecper $\leq \pi_1$)	0.006 (0.000)***	0.006 (0.000)***	0.001 (0.000)***	0.379 (0.027)***	0.078 (0.004)***		
($\pi_1 < \text{Tecper} \leq \pi_2$)	0.009 (0.000)***	0.008 (0.001)***	0.002 (0.000)***	0.431 (0.027)***	0.096 (0.004)***		
(Tecper $> \pi_2$)	0.014 (0.000)***	0.013 (0.001)***	0.005 (0.000)***	0.515 (0.028)***	0.124 (0.004)***		
Interm. No Thresh.						0.050 (0.003)***	0.050 (0.003)***
ln K goods (-1)							
(Tecper $\leq 0.07\%$)							0.008 (0.003)**
($0.07\% < \text{Tecper} \leq 0.94\%$)							0.017 (0.004)***
(Tecper $\leq 0.94\%$)						0.012 (0.003)***	
(Tecper $> 0.94\%$)						0.029 (0.004)***	0.030 (0.004)***
p-value (threshold)	0	0	0	0	0	0	.059
Observations	146,862	146,862	146,862	146,862	146,862	146,862	146,862
Firm fixed-effects	YES	YES	YES	YES	YES	YES	YES
Time fixed-effects	YES	YES	YES	YES	YES	YES	YES
F-stat	348.5	107.4	45.01	130	380	125.2	116.5
R-squared	0.40	0.31	0.25	0.40	0.44	0.40	0.40

*** p<0.01, ** p<0.05, * p<0.1. Standard errors in parentheses. All models estimated with time and firm-specific fixed-effects. Control variables omitted are the same as in Table 2. We add one to the log of Exports, Imports and the number of country-product varieties to avoid losing observations for firms that do not trade. in columns (1), (4) and (5) $\pi_1=0.07\%$ and $\pi_2=1.18\%$ in column (2) $\pi_1=0.07\%$ and $\pi_2=1.33\%$ in column (3) $\pi_1=0.13\%$ and $\pi_2=1.08\%$

sell higher-priced goods to wealthier destinations (Bastos et al., 2018; Manova & Zhang, 2012), and that developed countries produce the higher quality inputs used by firms for upgrading their products, while imports from developing countries tend to be connected to cost-saving effects (Bas & Strauss-kahn, 2015; Bas & Strauss-Kahn, 2014; Feng et al., 2016; Lo Turco & Maggioni, 2013). Therefore, the different origin/destination of inputs can be used as a proxy for the quality/technology embodied in these goods, which is indeed a common procedure in the literature.

Although our results only reveal minor differences regarding the effect of OECD imports, we do find a non-significant effect for the *Tecper* coefficient for non-OECD exports and a lower effect of non-OECD imports for higher capability firms and OECD exports in column (3). Therefore, the evidence of complementarity in the last two columns exists but appears weaker when compared to columns (1) and (2). This is likely related to the fact that dividing products according to destination does not account for vertical differentiation among products, a disadvantage that might be exacerbated by the export specialization pattern of the Brazilian industry, which to a great extent relies on regional (non-OECD) markets to sell some of its more technologically advanced products.

5.6. Extensions: country-product varieties and capital goods

We expand our framework to look more closely at the introduction of product-country varieties by firms. The relative importance of cost-competitiveness versus quality/technology might vary depending on

whether the firm is already a known player in a market or if it is an entrant trying to establish a reputation and new commercial relationships (Feng et al., 2016). Entrants might be more dependent on cost advantages to challenge established competitors and persuade buyers to change their suppliers, or alternatively, they might use innovation and quality to get a competitive edge over their competitors.

In Column (1) of Table 7, we replace the value of exports by the number of country-product pairs exported by firms, an indicator that firms are introducing new products or expanding into new markets, which will frequently involve adaptations, while in columns (2) and (3) we divide the number of country-product varieties into customized and non-customized products. Our results are similar to those reported for exported values in Tables (2) and (4) in the sense that we see a much stronger effect of imports for exports of customized products, and this effect increases at higher levels of technological capabilities. However, an important difference is in the fact that this increase appears to happen much faster in the case of exported country-product varieties. We see this result as a confirmation that quality is less important for lower capability firms to expand to new markets, whereas more capable firms rely in higher levels of innovativeness to expand the number of country-product pairs they sell, especially in customized goods.

In columns (4) and (5), we replace the value of imports by the logarithm of the number of imported product-country varieties of intermediates and look at their effect on the value and on the number of exported product-country pairs. Results demonstrate once again the presence of complementarities in both cases. A 1% increase in the number of imported country-product pairs is associated with an increase

of 0.51% in exported value and a 0.12% increase in the number of country-product pairs served by the more technologically capable firms. This is a sizeable effect and qualifies previous results that demonstrated the importance of (new) imported varieties but ignored asymmetries in technological capabilities between firms (Bas & Strauss-Kahn, 2014; Castellani & Fassio, 2019).

Finally, we consider the influence of imported capital goods on the export performance of local firms and possible complementarities with their technological capabilities. Capital goods are connected to investments in tangible assets and in the implementation of new production techniques. Imported capital goods, in particular, should embody frontier foreign technology and relate more closely with the implementations of process innovations by firms (Dosi et al., 2015; Hall, Ae, & Mairesse, 2009). In “technology gap” theories of development, they are connected to the ‘advantage of backwardness’ (Gerschenkron, 1962), i. e. the possibility that developing countries could benefit from the backlog of technologies developed previously in advanced nations. However, it is usually recognized that this knowledge can only be successfully applied by firms in developing countries if they already possess a minimum level of absorptive capacities (Abramovitz, 1989; Cohen & Levinthal, 1989; Verspagen, 1991).

Columns (5) and (6) show estimations for one and two thresholds for

imported capital goods, controlling for imports of intermediates. First, we find that the effect of capital goods remains considerably below that of imported intermediates at all levels of *Tecper*. Second, we find that there are also complementarities with technological capabilities, but they appear to be smaller: only one threshold is significant at the 5% level, while the most likely candidate for a second threshold is found to be not significant at this level. Firms with high technological capabilities indeed benefit more from imported capital goods than firms at lower levels as predicted by theory. However, complementarities appear much stronger for the case of intermediate products. Castellani & Fassio (2019) found a stronger effect for intermediates compared to capital goods and attributed the result to the possibility that intermediate goods embody technology and cost advantages to firms related to more input varieties, whereas capital goods only embody new technologies. Our results suggest that these effects are intensified for more capable firms, as complementarities are also a factor increasing the effect of intermediates, a pattern that is less present in the case of capital goods.

5.7. Robustness: Different quality measures and threshold variables

In Table 8 we perform a number of tests regarding our main choices of variables. We start by experimenting with different methods for

Table 8
Robustness 1: different quality measures and threshold variables.

DEP VARIABLE	(1)	(2)	(3)	(4)	(5)	(6)	(7)
THRESHOLD VARIABLE	Ln Exports	Ln Exports	Ln Exports	Ln Exports	Ln Exports	Ln Exports	Ln Exports
QUALITY VARIABLE	Tecper	Tecper	Tecper	College Per.	College Per.	TFP	TFP
	2	3	1	1	1	1	1
Tecper (-1)	2.34 (0.843)***	2.256 (0.840)***	3.215 (0.858)***	2.84 (0.687)***	2.816 (0.687)***	2.997 (0.687)***	2.960 (0.686)***
ln IM (-1)							
(Thresh $\leq \pi 1$)				0.043 (0.003)***		0.075 (0.008)***	
($\pi 1 < \text{Thresh} \leq \pi 2$)				0.056 (0.004)***		0.044 (0.003)***	
(Thresh $> \pi 2$)				0.076 (0.005)***		0.065 (0.004)***	
ln IM_HQ (-1)							
(Thresh $\leq \delta 1$)	0.015 (0.006)***	0.004 (0.006)***			0.023 (0.003)***		0.600 (0.006)***
($\delta 1 < \text{Thresh} \leq \delta 2$)	0.041 (0.007)***	0.032 (0.007)***			0.052 (0.005)***		0.023 (0.003)***
(Thresh $> \delta 2$)	0.06 (0.008)***	0.048 (0.008)***			0.062 (0.006)***		0.056 (0.005)***
ln IM_LQ (-1)							
(Thresh $\leq \lambda 1$)	0.033 (0.005)***	0.044 (0.005)***			0.028 (0.004)***		0.009 (0.017)
($\lambda 1 < \text{Thresh} \leq \lambda 2$)	0.02 (0.007)***	0.03 (0.006)***					0.056 (0.009)***
(Thresh $> \lambda 2$)	0.027 (0.008)***	0.043 (0.008)***			0.037 (0.005)***		0.037 (0.003)***
Avg Quality Int. (-1)							
(Thresh $\leq \Omega 1$)			-0.035 (0.010)***				
($\Omega 1 < \text{Thresh} \leq \Omega 2$)			0.002 (0.012)				
(Thresh $> \Omega 2$)			0.048 (0.015)***				
p-value (threshold)	0.17	0.06	0	0	0.11	0	0
Observations	125,176	125,176	125,176	146,862	125,176	146,862	125,176
Firm fixed-effects	YES	YES	YES	YES	YES	YES	YES
Time fixed-effects	YES	YES	YES	YES	YES	YES	YES
F-statistic	34.98	34.91	39.63	46.5	42.9	45	41
R-squared	0.40	0.41	0.40	0.39	0.38	0.40	0.39

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors in parentheses. Control variables omitted are the same as in Table 2, except where stated otherwise. We add one to Exports and Imports for the log transformation.

in columns (1) and (2): $\delta 1 = 0\%$, $\delta 2 = 1.18\%$, $\lambda 1 = 0.54\%$ and $\lambda 2 = 1.78\%$

in column (3) and (4): $\Omega 1 = 0\%$ and $\Omega 2 = 1.78\%$; $\pi 1 = 3.6\%$ and $\pi 2 = 6.2\%$

in column (5) and (6): $\delta 1 = 6.2\%$, $\delta 2 = 9.4\%$, and $\lambda 1 = 5.4\%$; $\pi 1 = 12.47\%$ and $\pi 2 = 14.6\%$

in column (7): $\delta 1 = 13\%$, $\delta 2 = 14.6\%$, $\lambda 1 = 11.27\%$ and $\lambda 2 = 12.47\%$

estimating quality. In column (1), we extract firm quality by estimating Eq. (14) using product-year (α_{pt}) and country fixed-effects (α_k) instead of country-year and product fixed effects, and call this quality measure 2. This option does not correspond as strictly to the demand function derived from the CES utility which we use throughout, but potentially controls better for product differences by creating year-specific product controls, which in turn will affect the classification of high and low-quality varieties. Nevertheless, column (1) shows very similar results compared to our main choice for quality estimation. In column (2) we use simple product-year averages of *unit prices* to classify imports in low and high-quality varieties (quality measure 3 in the table) and, once again, obtain very similar results. Finally, in column (3) we estimate Eq. (12) and (13) including directly the weighted average of imported intermediate quality at the firm-level in the model. For firms that do not import, we input zero quality to avoid losing these observations, which corresponds to implicitly assuming the quality of local inputs is slightly below the median quality of imported intermediates. Once again, we observe important complementarities between quality and firm capability.

Next, we test the sensitivity of our results to a different proxy for firms' technological capabilities. In columns (4) and (5), we use the participation of personnel with a college degree in the total labour force. This indicator is constructed from the same labour market data we used to obtain the *Tecper* variable but significantly less skewed, with the vast majority of the firms registering a positive value. We consider it to be a less precise indicator because it does not provide information about existing activities and functions within firms, despite being reasonably correlated to *Tecper*. Both columns (4) and (5) provide estimates compatible with the previous results and confirm the presence of the complementarities observed using the *Tecper* variable. The main difference is in Column (5), where we obtain a p-value of 11% for a second non-linearity in high-quality imports, therefore indicating the best model has only one threshold for this variable. Rather than at odds, this fact further confirms our initial impression that *Tecper* is indeed the better choice to capture firms' technological capabilities.

In columns (6) and (7), we replace *Tecper* with total factor productivity as the capability indicator. In the theoretical literature, productivity and capability are usually treated as the same variable, and that is also the case in the [Kugler & Verhoogen \(2012\)](#) model we explored in Section 2. In reality, firm-level productivity is the outcome of the interaction of several internal capabilities and external factors that influence firms' overall efficiency.¹⁷ Consequently, if complementarities with the quality of inputs are essentially derived from firms' technological capabilities, there is less reason to expect that productivity and higher quality inputs will present the same level of complementarity. This is exactly what we observe in columns (6) and (7). There are non-linearities in the effect of imported intermediates along the distribution of productivity, but these do not depict a discernable pattern that could be related to complementarities of any sort. Once we control for several correlated factors, including technological capabilities, it is not clear if the effect of imported intermediates on exports is growing or not for more productive firms, with results varying at different points in the distribution of technological capabilities in a haphazard manner.

In Table 9, we perform tests with different control variables. In columns (1) and (2) we include local intermediates to account for possible production scale effects not captured by other controls, whereas in columns (3) and (4) we exclude TFP from the models, both to exclude the possibility of controlling for firm capability twice and to avoid possible bias resulting from including an estimated variable among the regressors. In all cases, our results are not affected by these choices. Finally, in columns (5) and (6) we follow [Dosi et al. \(2015\)](#) and replace

TFP and schooling with value-added per employee and average wages paid. This framework is intended to also control for cost competitiveness and production efficiency, although the wage variable usually captures the quality of labour together with its cost. Once again, our results are not affected significantly by the change.

6. Conclusion

The economic literature has indicated that imported intermediates help firms to obtain cheaper and more varied inputs, frequently of higher quality and with superior embodied technology. In doing so, they allow them to produce more efficiently and increase the variety and quality of their products. Lower prices and better products, in turn, make traders more competitive in foreign markets, boosting exports in existing destinations and frequently helping them to open new ones. Evidence has confirmed that productivity and exports are positively affected by firms' increased use of imported inputs.

Our objective in this paper has been to investigate how innovation and firms' technological capabilities affect the relationship between imported intermediates and firms' export performance. We are motivated by the observation that the process by which firms transform new and better inputs into improved or new products is a process of technological change, and as such, is expected to depend on their skills to manage and implement innovations. Our analysis benefited greatly from datasets containing information about the functions performed by workers, which allowed us to create a proxy indicator of firms' technological capabilities. Using threshold regression techniques, we observed that the effect of imported intermediates on firms' export performance is positively affected by technological capabilities. These complementarities were found to be non-linear however, slowing at higher levels of capabilities, explaining why they could not be captured by simple models with linear interactions.

We found that these complementarities are exclusively found in the case of higher quality imports, indicating more capable firms benefit more from superior inputs to improve their exports. Lower quality imports were frequently found to be less important for firms with greater technological capabilities.

The type of exported product also appears to play a role in two ways. Firstly, there is a direct effect of technological capabilities on firms' export performance, but this is not relevant for products with a low scope for quality differentiation, while for products with a higher scope for differentiation the effect appears to be stronger. This result qualifies previous findings regarding the relationship between innovation and exports, which tend to emphasize the role of innovation irrespective of the characteristics of products firms are supplying. [Dosi et al. \(2015\)](#) highlight different sectoral patterns, but our results indicate transversal patterns that are only partially captured in sectoral divisions. It appears that for many exported products, innovation is not a relevant factor simply because there are low technological opportunities for firms to compete through product differentiation or because consumers are less inclined to pay for quality. Secondly, exporting products with a higher scope for quality differentiation intensifies complementarities between firm capability and input quality. This result is related to the previous one, as it indicates that complementarities become more important when opportunities for product innovation increase. Both of these patterns provide evidence in favour of theories that suggest the existence of complementarities between input quality and firm capability in the production of higher quality goods and with studies that emphasize the importance of technological capabilities for firms to successfully innovate.

This work also goes beyond firm heterogeneity in terms of productivity and points to innovation as an important determinant of firms' performance in external markets. We make explicit the role of internal capabilities connected to the innovation process, demonstrating both their direct and indirect influence on the international competitiveness of firms. We consider that this provides further encouragement for the

¹⁷ Ones needs to go no further than [Lall's \(1992\)](#) rich characterization of firm-level and national technological capabilities to have an example of this argument.

Table 9

Robustness 2: Different controls.

DEP VARIABLE CONTROL VARIABLES	(1) Ln Exports Including Local Intermediates	(2) Ln Exports Including Local Intermediates	(3) Ln Exports Excluding TFP	(4) Ln Exports Excluding TFP	(5) Ln Exports Including VAE and Wage	(6) Ln Exports Including VAE and Wage
Tecper (-1)	1.986 (0.697)***	2.300 (0.841)***	2.145 (0.704)***	2.515 (0.847)***	2.528 (0.761)***	2.627 (0.946)***
ln IM (-1) (Thresh $\leq \pi_1$)	0.044 (0.005)***		0.046 (0.005)***		0.047 (0.005)***	
($\pi_1 < \text{Thresh} \leq \pi_2$)	0.058 (0.006)***		0.061 (0.006)***		0.060 (0.006)***	
(Thresh $> \pi_2$)	0.072 (0.006)***		0.075 (0.007)***		0.074 (0.007)***	
ln IM_HQ (-1) (Thresh $\leq \delta_1$)		0.015 (0.006)***		0.017 (0.006)***		0.018 (0.006)***
($\delta_1 < \text{Thresh} \leq \delta_2$)		0.043 (0.007)***		0.045 (0.007)***		0.044 (0.007)***
(Thresh $> \delta_2$)		0.062 (0.008)***		0.063 (0.008)***		0.063 (0.008)***
ln IM_LQ. Int. (-1) (Thresh $\leq \lambda_1$)		0.037 (0.005)***		0.038 (0.005)***		0.037 (0.006)***
($\lambda_1 < \text{Thresh} \leq \lambda_2$)		0.022 (0.007)***		0.023 (0.007)***		0.023 (0.007)***
(Thresh $> \lambda_2$)		0.031 (0.008)***		0.032 (0.008)***		0.032 (0.008)***
ln VAE					0.088 (0.014)***	0.075 (0.015)***
ln Wage					0.024 (0.032)	-0.007 (0.032)
p-value (threshold)	0	0.1	0	0.09	0	0.09
Observations	146,854	125,246	146,947	125,248	141,383	120,802
Firm fixed-effects	YES	YES	YES	YES	YES	YES
Time fixed-effects	YES	YES	YES	YES	YES	YES
F-statistic	54.66	34.12	54.13	33.83	50.17	29.77
R-squared	0.39	0.41	0.38	0.39	0.39	0.40

*** p<0.01, ** p<0.05, * p<0.1. Standard errors in parentheses. Control variables omitted are the same as in Table 2, except where stated otherwise. We add one to Exports and Imports for the log transformation.

in column (1): $\pi_1=0.07\%$ and $\pi_2 = 1.08\%$

in column (2): $\delta_1=0\%$, $\delta_2=1.18\%$, $\lambda_1=0.54\%$ and $\lambda_2=1.78\%$

in column (3): $\pi_1=0.07\%$ and $\pi_2 = 1.16\%$

in column (4): $\pi_1=0.07\%$ and $\pi_2 = 1.18\%$

in column (6): $\delta_1=0.06\%$, $\delta_2=1.18\%$, $\lambda_1=0.54\%$ and $\lambda_2=1.78\%$

exploration of empirical and theoretical approaches that, although not incompatible, go beyond the productivity sorting mechanism described in Melitz-type models of international trade, and look at innovation as one of the main determinants of firms' external performance. Among other benefits, such approaches reveal other dimensions by which trade can affect firms in asymmetrical ways, possibly providing valuable inputs for public policy. In that regard, we showed here that the benefits of accessing intermediates – for instance, in episodes of trade liberalization – will tend to be maximized in the subgroup of highly capable firms, whereas low-skilled firms might not have the ability to take full advantage of the enhanced quality and variety provided by imports.

Appendix I. Descriptive statistics

Tables A1 and A2.

Credit author statement

Caio Torres Mazzi: Conceptualization; Methodology; Investigation; Formal Analysis; Writing

Neil Foster-McGregor: Methodology; Formal Analysis; Writing; Supervision

Declaration of Competing Interest

None.

Table A1

Number, average value and country-product pairs of exporters and importers.

		2001	2002	2003	2004	2005	2006	2007
Exporters	Firms	6,979	7,000	7,279	7,764	7,724	8,127	8,062
	Value (mean, US\$ 1000)	6,174	6,678	7,970	9,786	11,994	13,269	15,610
	Country-prod. varieties (mean)	25.01	26.4	29.01	31.29	34.58	36.25	38.1
Importers of Interm.	Firms	6,503	6,306	6,137	6,308	6,529	6,964	7,316
	Value (mean, US\$ 1000)	4,894	4,125	5,015	6,691	7,312	8,291	10,121
	Country-prod. varieties (mean)	29.01	29.36	30.99	32.32	32.82	32.87	33.74
Total Firms		25,237	26,613	27,986	28,788	30,658	32,077	31,517

Table A.2

Descriptive statistics of the main variables according to the presence of technical personnel in the firm, means and standard deviations.

		ln TFP	ln EMP	ln K/L	Age	Schooling	Tecper (%)	College deg. (%)
Tecper	> 0	12.62 (1.28)	3.94 (0.69)	6.52 (5.68)	12.90 (9.16)	7.930 (1.76)	0 (0.036)	2.0% (0.05)
	= 0	14.13 (1.51)	5.07 (1.21)	10.25 (3.92)	21.41 (12.66)	9.220 (1.90)	2.3% (0.036)	9.0% (0.11)
		ln EX (R\$)	ln IM (R\$)	ln IM_HQ (R\$)	ln IM_LQ (R\$)	ln # country-prod imported	ln # country-prod exported	
Tecper	> 0	2.15 (4.98)	1.45 (4.07)	1.01 (3.47)	1.01 (3.33)	0.28 (0.75)	0.33 (0.82)	
	= 0	7.9 (7.73)	7.58 (7.45)	6.47 (7.30)	6.09 (6.80)	2.32 (1.53)	1.44 (1.67)	

Appendix II. Threshold regression model

We follow closely the procedures described by Hansen (1999) and employed by Falvey et al (2006). Initially, we estimate the model with a single threshold for imported intermediates as follows:

$$\ln EX_{it} = \beta_0 + \delta_1 \ln IM_{it-1} I(Tecper_{it-1} < \lambda_1) + \delta_2 \ln IM_{it-1} I(Tecper_{it-1} \geq \lambda_1) + \beta_1 \ln \left(\frac{K}{L} \right)_{it-1} + \beta_2 \ln EMP_{it-1} + \beta_3 SCHOOL_{it-1} + \beta_4 \ln TFP_{it-1} + \beta_5 Tecper_{it-1} + \alpha_i + \nu_i + \mu_{it} \quad (A1)$$

where δ_1 and δ_2 are the coefficients on $\ln IM$ in the low and high regime as defined by the threshold value, λ_1 of *Tecper*. We estimate λ_1 by least squares, which involves: (i) estimating Eq. (A1) for distinct values of λ_1 ; and (ii) obtaining an estimate of $\hat{\lambda}_1$ by selecting the value of λ_1 that minimizes the sum of squared errors obtained from the regressions in step (i). We limit our search to values of *Tecper* below the 90th percentile,¹⁸ and define a minimum range of 10% for each regime, i.e. each estimated regime must include at least 10% of observations within the estimation interval. Once we obtain $\hat{\lambda}_1$, estimation of $\hat{\lambda}_2$ is obtained by the same procedure, i.e. choosing the values of λ_1 and λ_2 that minimize the sum of squared errors of the following equation:

$$\ln EX_{it} = \beta_0 + \delta_1 \ln IM_{it-1} I(Tecper_{it-1} \leq \lambda_1) + \delta_2 \ln IM_{it-1} I(\lambda_1 < Tecper_{it-1} \leq \lambda_2) + \delta_3 \ln IM_{it-1} I(Tecper_{it-1} > \lambda_2) + \beta_1 \ln \left(\frac{K}{L} \right)_{it-1} + \beta_2 \ln EMP_{it-1} + \beta_3 SCHOOL_{it-1} + \beta_4 \ln TFP_{it-1} + \beta_5 Tecper_{it-1} + \alpha_i + \nu_i + \mu_{it} \quad (A2)$$

Since this is computationally demanding, we follow a sequential procedure, i.e. we fix $\hat{\lambda}_1$ while searching for $\hat{\lambda}_2$ according to steps (i) and (ii) above. Adopting this approach, $\hat{\lambda}_2$ is efficient but $\hat{\lambda}_1$ is not. We therefore follow the suggestion of Hansen (1999) and take a third step, namely: (iii) fix $\hat{\lambda}_2$ and search once again for the first threshold, which we call $\hat{\lambda}_1^r$.

At each step we estimate the significance of the threshold we obtain. To do this, it is necessary to follow a bootstrap procedure since under the null hypothesis – $H_0 : \delta_1 = \delta_2$ for (A1) or $H_0 : \delta_2 = \delta_3$ for (A2) – the threshold is not identified and the test has a non-standard distribution. We once again follow the approach of Hansen (1999) and bootstrap to simulate the distribution of the likelihood ratio (F_1) test and obtain a p-value for the test. The likelihood ratio test is given by the following equation:

$$F_1 = n(t-1) \frac{S_0 - S_1}{S_1}$$

where S_0 and S_1 are the residual sum of squares under the null and alternative hypotheses, respectively. The bootstrap is created by drawing from the distribution of the residuals of the estimated threshold model and repeating 1,000 times. The p-value is the number of times the simulated F_1 is above the actual one and we adopt a 5% level of significance for rejecting the null.

We adopt an analogous procedure to estimate the models with the two non-linear variables, high and low-quality imported intermediates: we

¹⁸ We tested for values of *Tecper* until the 95th percentile in initial estimations of (A1) but there was no improvement in the quality of adjustment after the 90th percentile.

follow steps (i) and (ii) for high-quality imports and repeat (i) and (ii) for low-quality imports fixing the thresholds found for the other variable. We then adapt (iii) for the two-variable case by fixing the number of thresholds found for low-quality imports and repeating (i) and (ii) for high-quality imports.

References

- Abramovitz, M., 1989. Thinking about growth and other essays on economic growth and welfare. Chap 1 Thinking about growth and other essays on economic growth and welfare, 1st edn. Cambridge University Press, New York, p. 406.
- Akerberg, D., Caves, K., & Frazer, G. (2006). Structural identification of production functions. In *mimeo*. <https://doi.org/10.3982/ECTA13408>.
- Amity, M., Konings, J., 2007. Trade liberalization, intermediate inputs, and productivity: evidence from Indonesia. *Am. Econ. Rev.* 97 (2003), 1611–1638.
- Anderson, J.E., Van Wincoop, E., 2004. Trade costs. *J. Econ. Lit.* 42 (3), 691–751. <https://doi.org/10.1257/0022051042177649>.
- Andersson, M., Weiss, J.F., 2012. External trade and internal geography: local export spillovers by industry characteristics and firm size. *Spat. Econ. Anal.* 7 (July 2013), 421–446. <https://doi.org/10.1080/17421772.2012.722664>.
- Antràs, P., Chor, D., 2013. Organizing the global value chain. *Econometrica* 81 (6), 2127–2204. <https://doi.org/10.3982/ECTA10813>.
- Antràs, P., Helpman, E., 2004. Global sourcing. *J. Pol. Econ.* 112 (3), 552–580. <https://doi.org/10.1086/383099>.
- Araújo, B.C., 2016. Efeitos de aprendizado de exportação: Diferenças quanto à permanência, destinos de exportação, tamanho e intensidade tecnológica. In: De Negri, F., Cavalcante, L.R. (Eds.), *Produtividade no Brasil Volume 2 - Determinantes*, 1st edn. Ipea, Brasília, pp. 153–171.
- Araújo, B.C., Cavalcante, L.R., Alves, P., 2009. Variáveis proxy para os gastos empresariais em inovação com base no pessoal ocupado técnico-científico disponível na Relação Anual de Informações Sociais (Rais). Ipea, Brasília. <http://repositorio.ipea.gov.br/handle/11058/3588>.
- Ariste, D., Castellani, D., Franco, C., 2013. Firms' exporting and importing activities: Is there a two-way relationship. *Rev. World Econ.* 149 (1), 55–84. <https://doi.org/10.1007/s10290-012-0137-y>.
- Aw, B.Y., Roberts, M.J., Winston, T., 2007. Export market participation, investments in R & D and worker training, and the evolution of firm productivity. *The World Economy* 30 (1), 83–104. <https://doi.org/10.1111/j.1467-9701.2007.00873.x>.
- Barrios, S., Gorg, H., Strobl, E., 2003. Explaining firms' export behavior: R&D, spillovers, and the destination market. *Oxford Bull. Econ. Stat.* 63 (4), 475–496.
- Bas, M., 2012. Input-trade liberalization and firm export decisions: Evidence from Argentina. *J. Dev. Econ.* 97 (2), 481–493. <https://doi.org/10.1016/j.jdevco.2011.05.010>.
- Bas, M., Strauss-kahn, V., 2015. Input-trade liberalization, export prices and quality upgrading. *J. Int. Econ.* 95 (2), 250–262. <https://doi.org/10.1016/j.jinteco.2014.12.005>.
- Bas, M., Strauss-Kahn, V., 2014. Does importing more inputs raise exports? Firm-level evidence from France. *Rev. World Econ.* 150 (2), 241–275. <https://doi.org/10.1007/s10290-013-0175-0>.
- Bastos, P., Silva, J., Verhoogen, E., 2018. Export destinations and input prices : evidence from Portugal. *Am. Econ. Rev.* 108 (2), 353–392. Retrieved from. http://wtorg.hs3esdk.ru/english/news_e/events_e/trade_workshop13_e/paulo_bastos_6_june_e.pdf.
- Becker, S.O., Egger, P.H., 2013. Endogenous product versus process innovation and a firm's propensity to export. *Emp. Econ.* 44, 329–354. <https://doi.org/10.1007/s00181-009-0322-6>.
- Bell, M., Pavitt, K., 1993. Technological accumulation and industrial growth: contrast between developed and developing countries. *Tech. Glob. Econ. Perform.* 2 (2), 157–210.
- Bell, M., Pavitt, K., 1995. The development of technological capabilities. In: ul Haque, I. (Ed.), *Trade, Technology and International Competitiveness*, 1, pp. 69–102. <https://doi.org/10.1080/758539006>.
- Bernard, A.B., Jensen, J.B., 1999. Exceptional exporter performance: cause, effect, or both. *J. Int. Econ.* 47 (1), 1–25. [https://doi.org/10.1016/S0022-1996\(98\)00027-0](https://doi.org/10.1016/S0022-1996(98)00027-0).
- Bernard, A.B., Jensen, J.B., 2004. Why some firms export. *Rev. Econ. Stat.* 86 (2), 561–569.
- Caldera, A., 2010. Innovation and exporting: evidence from Spanish manufacturing firms. *Rev. World Econ.* 146 (4), 657–689. <https://doi.org/10.1007/s10290-010-0065-7>.
- Cassiman, B., Golovko, E., Martínez-Ros, E., 2010. Innovation, exports and productivity. *Int. J. Indust. Org.* 28 (4), 372–376.
- Castellani, D., Fiasio, C., 2019. From new imported inputs to new exported products. Firm-level evidence from Sweden. *Research Policy* 48 (1), 322–338. <https://doi.org/10.1016/j.respol.2018.08.021>.
- Clerides, S., Lach, S., Tybout, J., 1998. Is “ Learning-by-exporting ” important? Micro-dynamic evidence from Colombia, Mexico and Morocco. *Quart. J. Econ.* 113 (3), 903–947.
- Cohen, W.M., Levinthal, D.A., 1989. Innovation and Learning: The Two Faces of R&D. *Econ. J.* 99 (397), 569–596.
- Damijan, J.P., Kostevc, Č., 2015. Learning from Trade through Innovation. *Oxford Bull. Econ. Stat.* 77 (3), 408–436. <https://doi.org/10.1111/obes.12071>.
- Del Prete, D., Rungi, A., 2015. *Organizing the Global Value Chain : a firm-level test* (No. 04). <https://doi.org/10.13140/RG.2.1.1826.1928>.
- Demir, B.F.B., 2013. Trading Intermediates and Quality.
- Dosi, G., Grazzi, M., Moschella, D., 2015. Technology and costs in international competitiveness: From countries and sectors to firms. *Research Policy* 44 (10), 1795–1814. <https://doi.org/10.1016/j.respol.2015.05.012>.
- Edwards, L., Sanfilippo, M., Sundaram, A., 2018. Importing and firm export performance: new evidence from South Africa. *South African J. Econ.* 86 (January 2016), 79–95. <https://doi.org/10.1111/saje.12154>.
- Falvey, R., Foster, N., Greenaway, D., 2006. Intellectual property rights and economic growth. *Rev. Dev. Econ.* 10 (4), 700–719.
- Fan, H., Li, Y.A., Yeaple, S.R., 2015. Trade liberalization, quality, and export prices. *Rev. Econ. Stat.* 97 (5), 1033–1051. https://doi.org/10.1162/REST_a_00524.
- Feng, L., Li, Z., Swenson, D.L., 2016. The connection between imported intermediate inputs and exports: Evidence from Chinese firms. *J. Int. Econ.* 101, 86–101. <https://doi.org/10.1016/j.jinteco.2016.03.004>.
- Flach, L., 2016. Quality upgrading and price heterogeneity: Evidence from Brazilian exporters. *J. Int. Econ.* 102, 282–290. <https://doi.org/10.1016/j.jinteco.2016.07.005>.
- Gerschenkron, A., 1962. *Economic Backwardness in Historical Perspective: A Book of Essays*. Belknap Press of Harvard University Press, Cambridge, MA.
- Goldberg, P.K., Khandelwal, A.M., Pavcnik, N., Topalova, P., 2010. Imported intermediate inputs and domestic product growth: evidence from India. *Quart. J. Econ.* 125 (4), 1727–1767.
- Greenaway, D., Kneller, R., 2007. Firm heterogeneity, exporting and foreign direct investment. *Econ. J.* 117 (1996), F134–F161. <https://doi.org/10.1111/j.1468-0297.2007.02018.x>.
- Greenaway, D., Sousa, N., Wakelin, K., 2004. Do domestic firms learn to export from multinationals. *Eur. J. Pol. Econ.* 20 (4), 1027–1043. <https://doi.org/10.1016/j.ejpoleco.2003.12.006>.
- Hall, B.H., Ae, F.L., Mairesse, J., 2009. Innovation and productivity in SMEs: empirical evidence for Italy. *Small Busi. Econ.* 33, 13–33. <https://doi.org/10.1007/s11187-009-9184-8>.
- Hallak, J.C., Sivadasan, J., 2013. Product and process productivity: Implications for quality choice and conditional exporter premia. *J. Int. Econ.* 91 (1), 53–67. <https://doi.org/10.1016/j.jinteco.2013.05.001>.
- Halpern, L., Koren, M., Szeidl, A., 2015. Imported inputs and productivity. *Am. Econ. Rev.* 105 (12), 3660–3703. <https://doi.org/10.1257/aer.20150443>.
- Hansen, B.E., 1999. Threshold effects in non-dynamic panels: Estimation, testing and inference. *J. Econometrics* 93, 345–368.
- Harrigan, J., Reshef, A., Toubal, F., 2016. The March of the Techies: Technology, Trade, and Job Polarization in France, 1994–2007. NBER Working Paper Series. <https://doi.org/10.1017/CBO9781107415324.004>.
- Johnson, R.C., 2011. Trade and prices with heterogeneous firms. *J. Int. Econ.* 86 (1), 43–56. <https://doi.org/10.1016/j.jinteco.2011.09.004>.
- Kannebley Jr, Sergio, S., Esteves, A., L., Silva, A.M.P.da, Araújo, B.C., 2009. Auto-seleção e aprendizado no comércio exterior das firmas industriais brasileiras. *Revista ANPEC* 10 (4), 715–740.
- Kasahara, H., Lapham, B., 2013. Productivity and the decision to import and export: Theory and evidence. *J. Int. Econ.* 89 (2), 297–316. <https://doi.org/10.1016/j.jinteco.2012.08.005>.
- Kasahara, H., Rodrigue, J., 2008. Does the use of imported intermediates increase productivity. *J. Dev. Econ.* 87, 106–118. <https://doi.org/10.1016/j.jdevco.2007.12.008>.
- Khandelwal, A., Schott, P. K., & Wei, S. (2013). Trade liberalization and embedded institutional reform : evidence from Chinese exporters. 103(6), 2169–2195.
- Kugler, M., Verhoogen, E., 2008. The quality-complementarity hypothesis: theory and evidence from Colombia (No. 14418). Cambridge, MA.
- Kugler, M., Verhoogen, E., 2009a. Plants and imported inputs : new facts and an interpretation. *Am. Econ. Rev. Paper Proc.* 99 (2), 501–507.
- Kugler, M., Verhoogen, E., 2012. Prices, plant size, and product quality. *Rev. Econ. Stud.* 79 (1), 307–339. <https://doi.org/10.1093/restud/rdr021>.
- Lall, S., 1992. Technological capabilities and industrialization. *World Dev.* 20 (2), 165–186. [https://doi.org/10.1016/0305-750X\(92\)90097-F](https://doi.org/10.1016/0305-750X(92)90097-F).
- Levinsohn, J., Petrin, A., 2003. Estimating production functions using inputs to control for unobservables. *Rev. Econ. Stud.* 70 (2), 317–341. <https://doi.org/10.1111/1467-937X.00246>.
- Lo Turco, A., Maggioni, D., 2013. On the role of imports in enhancing manufacturing exports. *The World Econ.* <https://doi.org/10.1111/j.1467-9701.2012.12020.x>.
- Manova, K., & Yu, Z. (2017). *Multi-product firms and product quality supplementary online appendix*. 109, 116–137. Retrieved from <https://ars.els-cdn.com/content/image/1-s2.0-S0022199617301101-mmc1.pdf>.
- Manova, K., Zhang, Z., 2012. Export prices across firms and destinations. *Quart. J. Econ.* 127 (1), 379–436. <https://doi.org/10.1093/qje/qjr051>.
- Melitz, M., 2003. The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica* 71 (6), 1695–1725. <https://doi.org/10.1007/BF01886181>.
- Messa, A., 2014. Metodologia de cálculo da produtividade total dos fatores e da produtividade da mão de obra. In: De Negri, F., Cavalcante, L.R. (Eds.), *Produtividade no Brasil : desempenho e determinantes*, pp. 87–110. Retrieved from. <http://www.ipea.gov.br>.

- Morrison, A., Pietrobelli, C., Rabellotti, R., 2008. Global value chains and technological capabilities : a framework to study learning and innovation in developing countries. *Oxford Dev. Stud.* 36 (1), 39–58. <https://doi.org/10.1080/13600810701848144>.
- Olley, B.Y.G.S., Pakes, A., 1996. The dynamics of productivity in the telecommunications equipment industry. *Econometrica* 64 (6), 1263–1297.
- Pisu, M., Muûls, M., 2009. Imports and exports at the level of the firm : evidence from Belgium. *The World Economy* 32 (5), 692–734. <https://doi.org/10.1111/j.1467-9701.2009.001172.x>.
- Rauch, J.E., 1999. Networks versus markets in international trade. *J. Int. Econ.* 48 (1), 7–35. [https://doi.org/10.1016/S0022-1996\(98\)00009-9](https://doi.org/10.1016/S0022-1996(98)00009-9).
- Roberts, M. J., & Tybout, J. R. (1999). An empirical model of sunk costs and the decision to export. 87(4), 545–564. <https://doi.org/10.1596/1813-9450-1436>.
- Topalova, P., Khandelwal, A., 2011. Trade liberalization and firm productivity: the case of India. *Rev. Econ. Stat.* 93 (3), 995–1009. https://doi.org/10.1162/REST_a_00095.
- Verspagen, B., 1991. A new empirical approach to catching up or falling behind. *Struct. Change Econ. Dyn.* 2 (2), 359–380. [https://doi.org/10.1016/S0954-349X\(05\)80008-6](https://doi.org/10.1016/S0954-349X(05)80008-6).
- Wagner, J., 2007. Exports and productivity: A survey of the evidence from firm-level data. *World Econ.* 30 (1), 60–82. <https://doi.org/10.1111/j.1467-9701.2007.00872.x>.
- Wagner, J., 2016. A survey of empirical studies using transaction level data on exports and imports. *Rev. World Econ.* 152 (1), 215–225. <https://doi.org/10.1007/s10290-015-0235-8>.